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Kriege

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(54) **CYLINDER FOR A SHEET-TREATING AND/OR SHEET-PROCESSING MACHINE WITH SUCTION AIR OPENINGS AND MACHINE FOR TREATING AND/OR PROCESSING SHEET-FORMAT SUBSTRATE COMPRISING SUCH A CYLINDER**

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(Continued)

(52) **U.S. Cl.**

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(58) **Field of Classification Search**

CPC **B41F 31/18**; **B41F 11/02**; **B41F 21/00**; **B65H 5/226**; **B65H 5/2406**; **B65H 5/332**;

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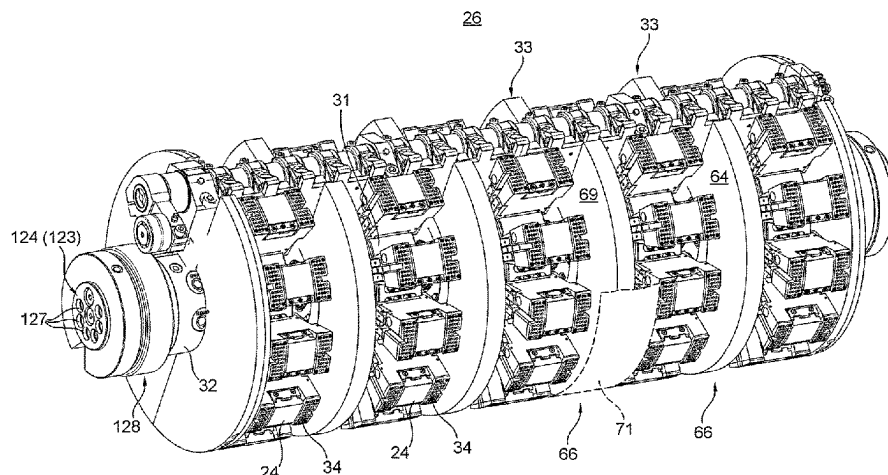
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(57)

ABSTRACT

Examples include a cylinder having a holding means around the circumference and by which a substrate sheet can be picked up and held during rotation of the cylinder over a rotation angle range between when the substrate sheet is received and is transferred downstream. A rotary union includes a rotor rotating along with the cylinder which, at an end face, is arranged with a stator that does not rotate during regular operation. A vacuum pressure can be applied via the rotary union to suction openings provided around the circumference of the cylinder while running through an active

(Continued)



rotation angle phase of the cylinder. The rotary union can be set in terms of a size of a passage angle sector determining the size of the active rotation angle phase and/or in terms of a position of the passage angle sector determining the position of the active rotation angle phase.

14 Claims, 18 Drawing Sheets

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B65H 5/22 (2006.01)

(52) **U.S. Cl.**

CPC *B65H 2406/332* (2013.01); *B65H 2801/21*
(2013.01)

(58) **Field of Classification Search**

CPC *B65H 2801/21*; *B65H 2406/33*; *B65H*
2406/3312

See application file for complete search history.

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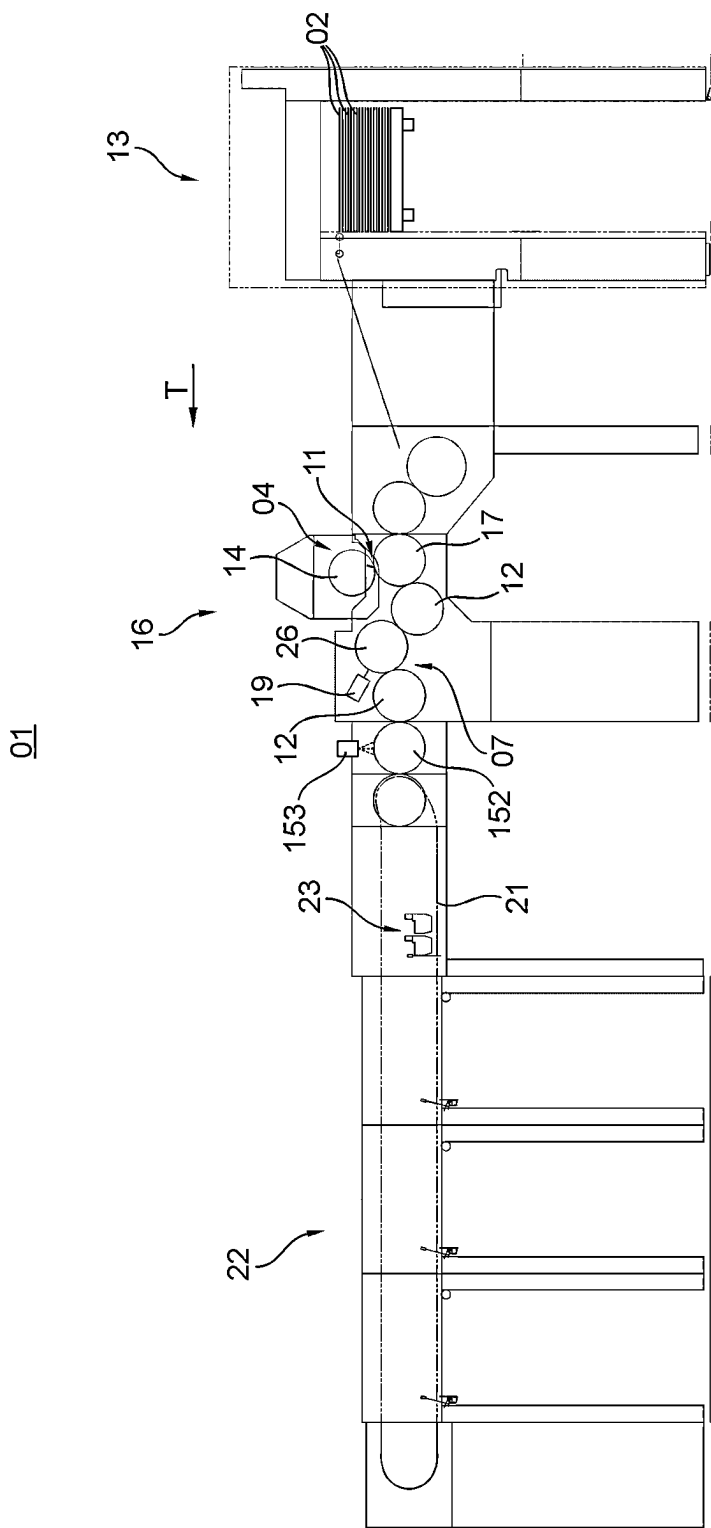


Fig. 1

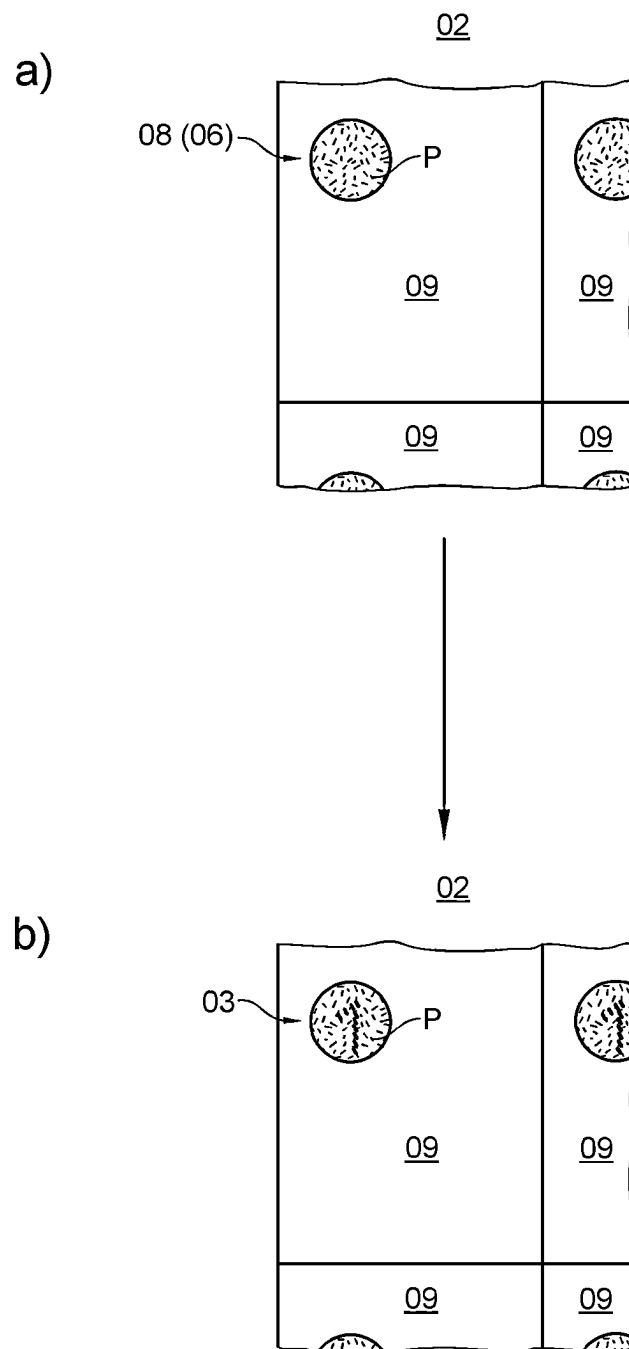


Fig. 2

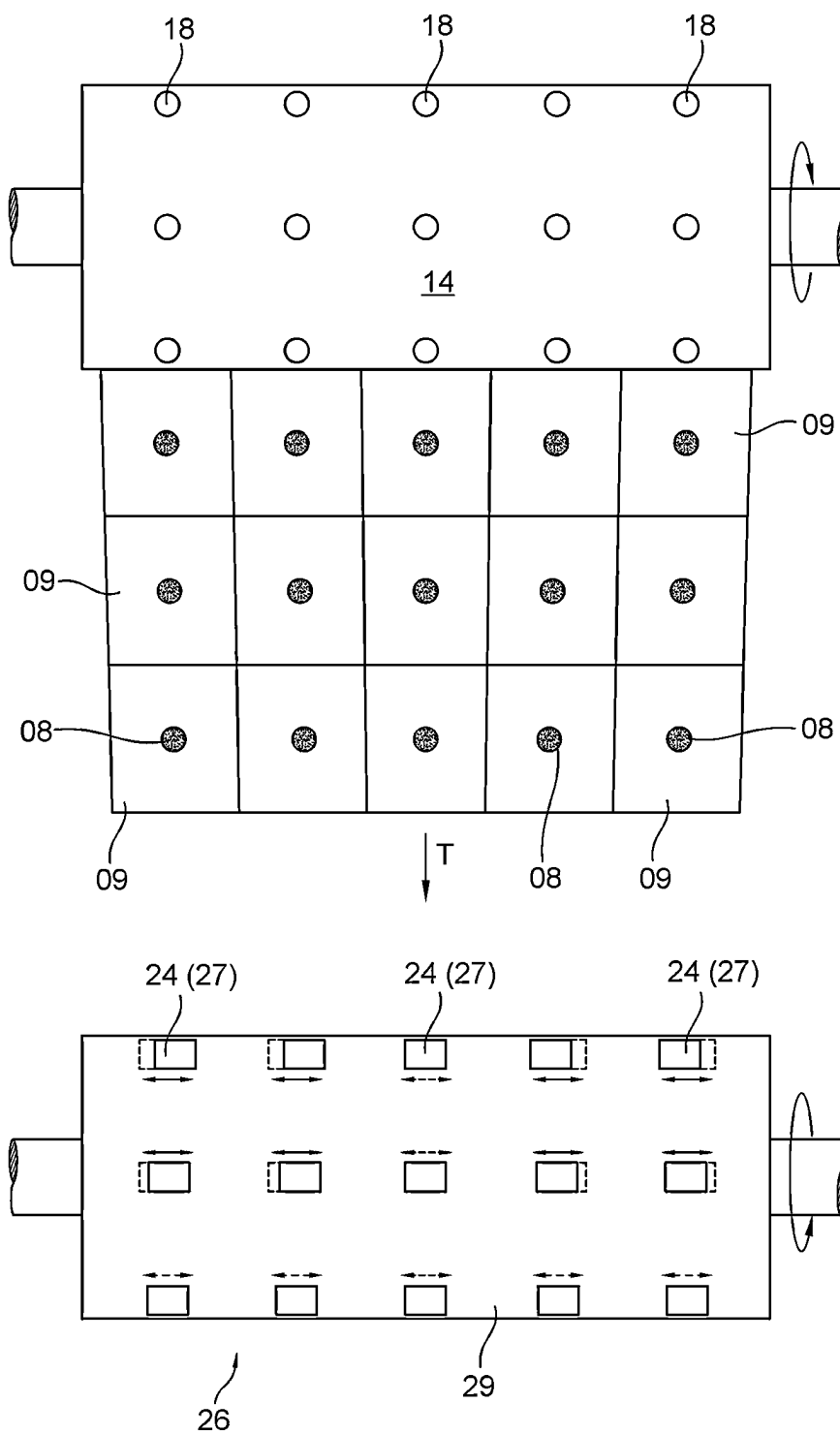


Fig. 3

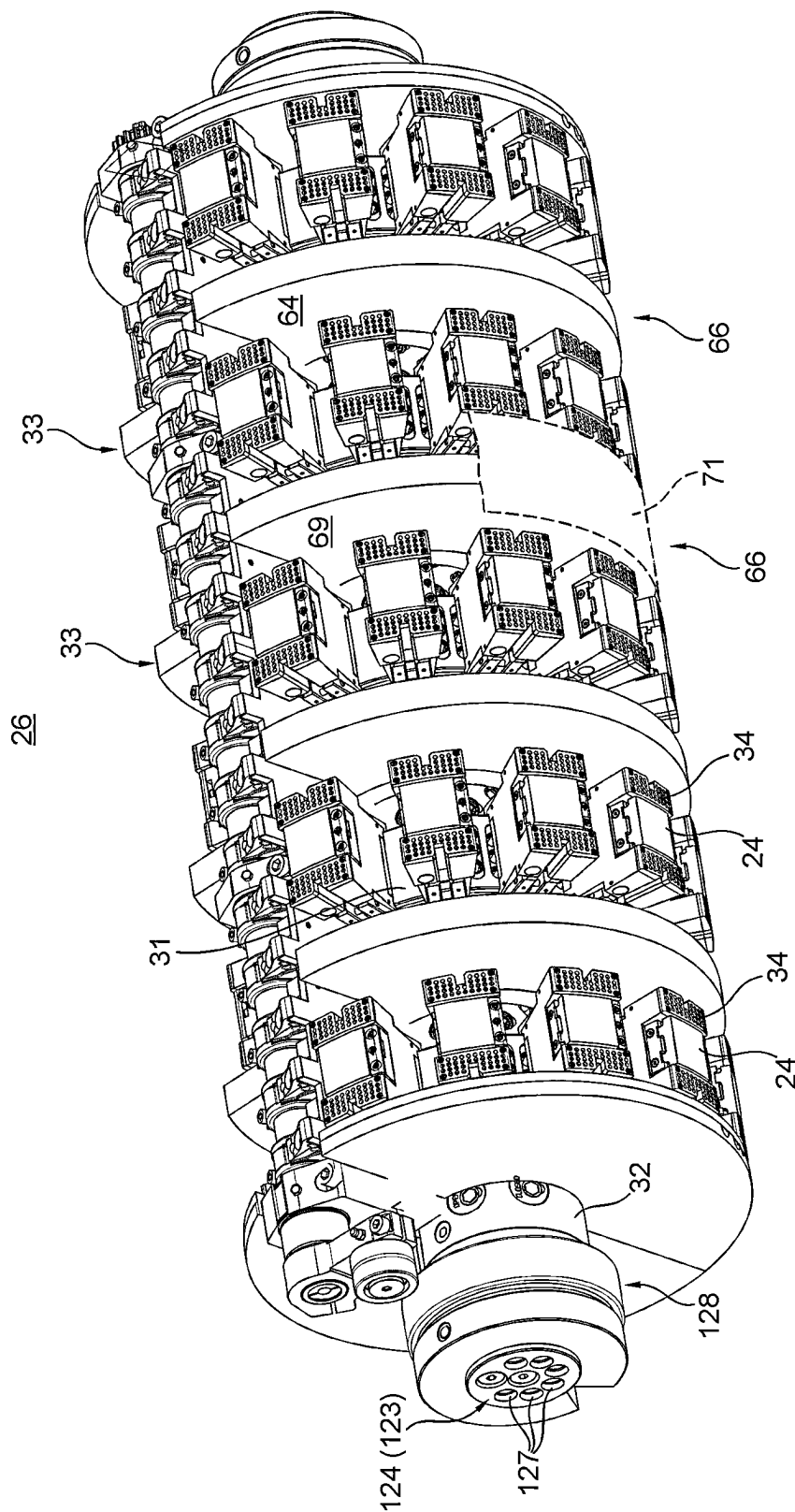


Fig. 4

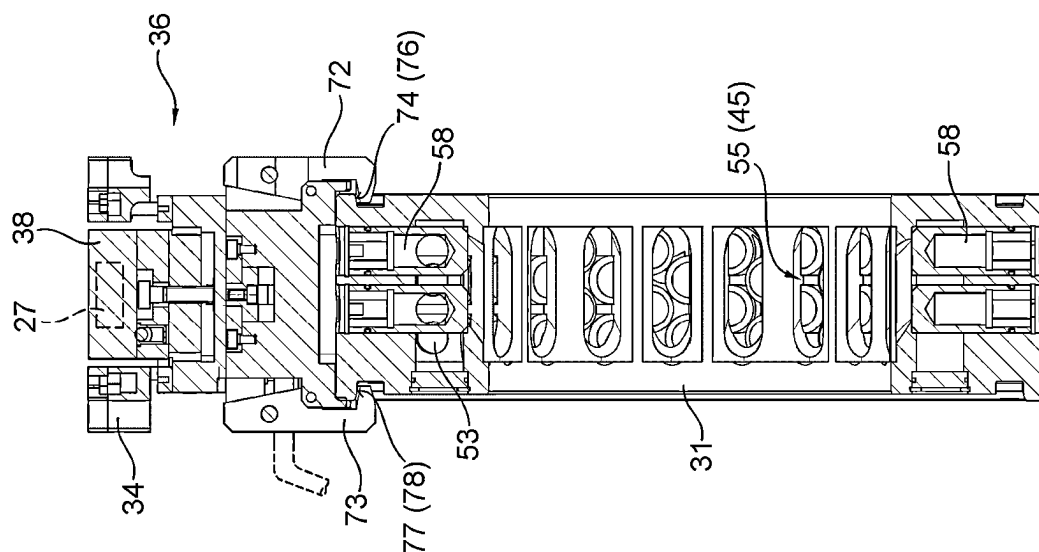


Fig. 6

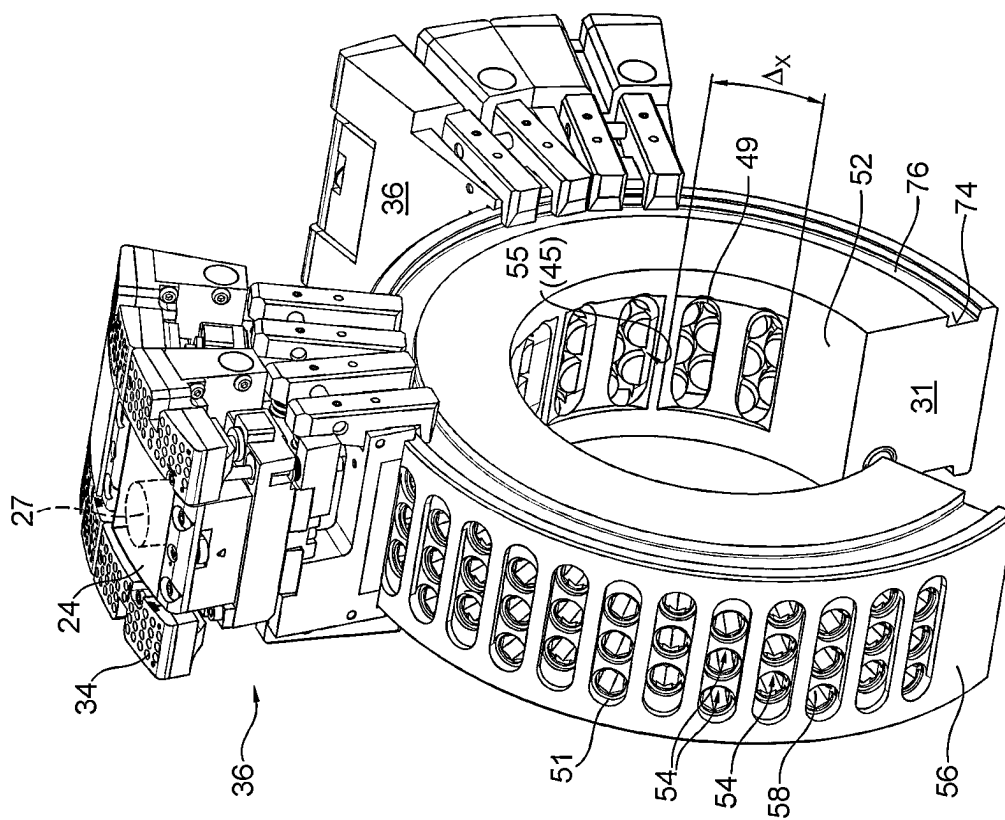


Fig. 5

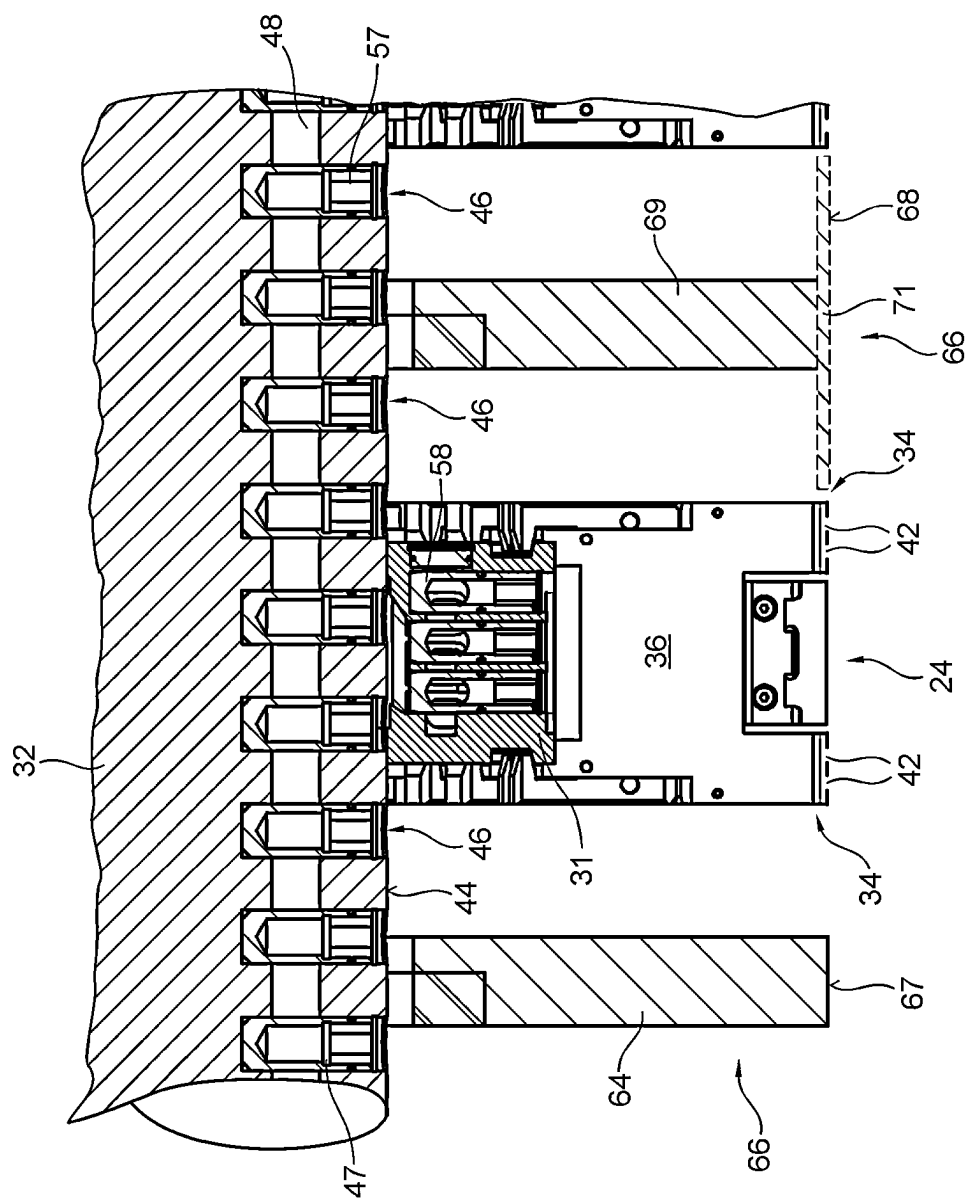


Fig. 7

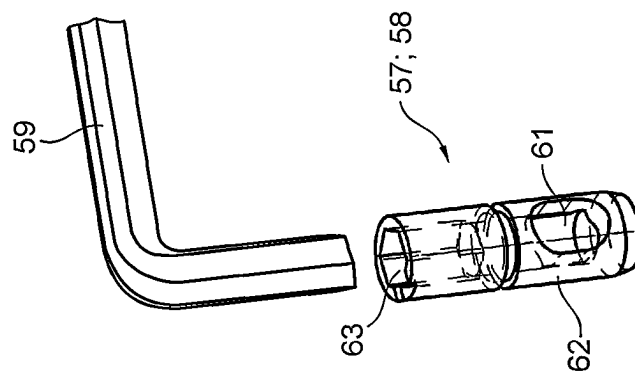


Fig. 8

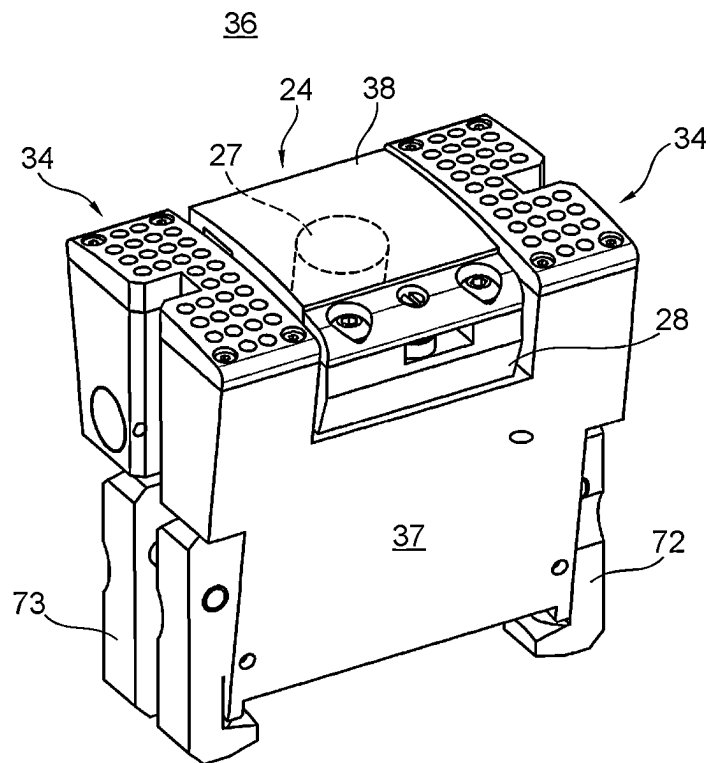


Fig. 9

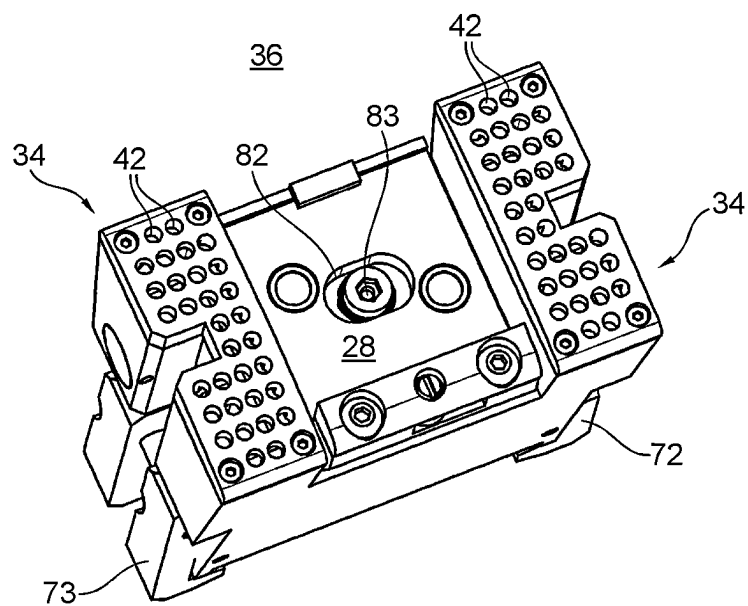


Fig. 10

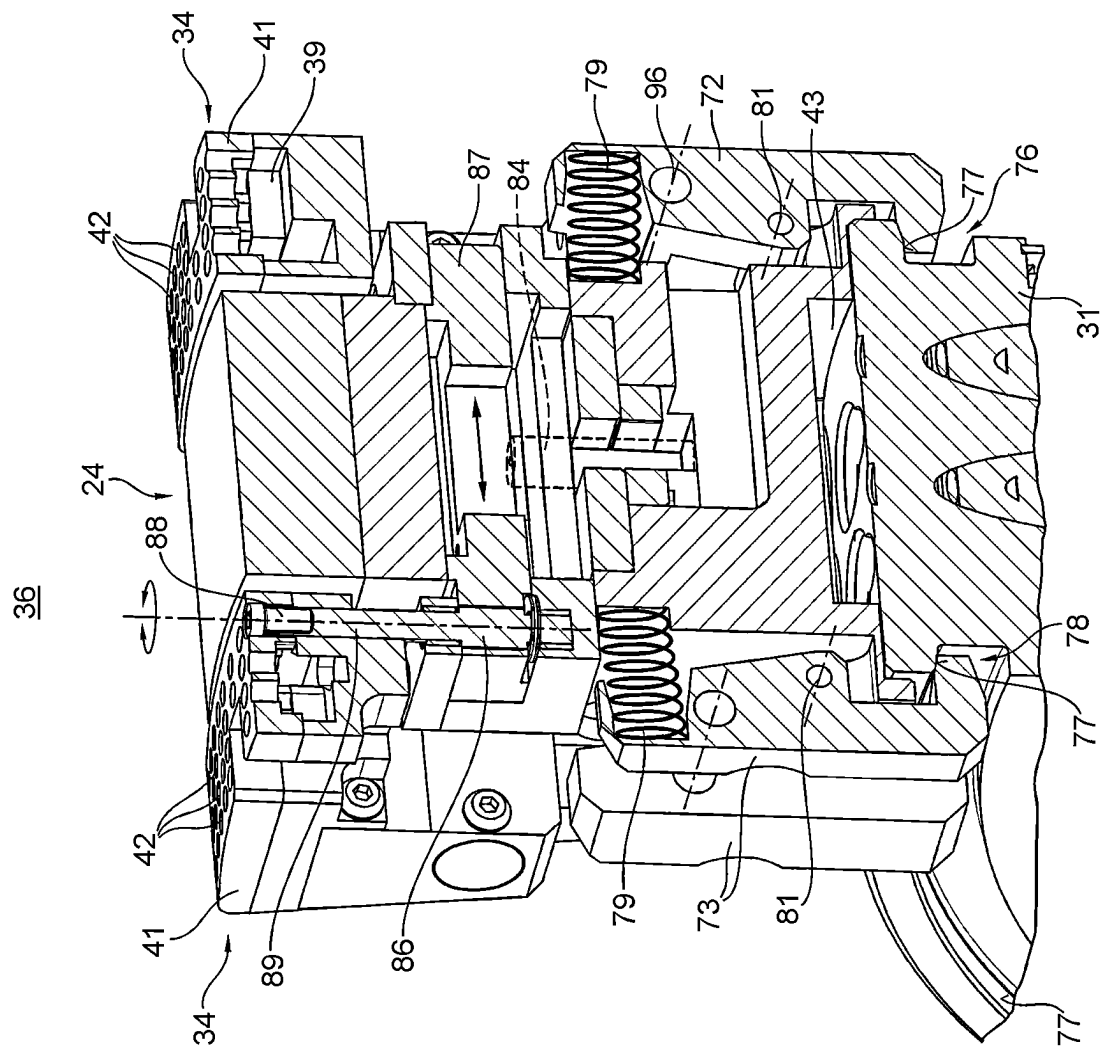


Fig. 11

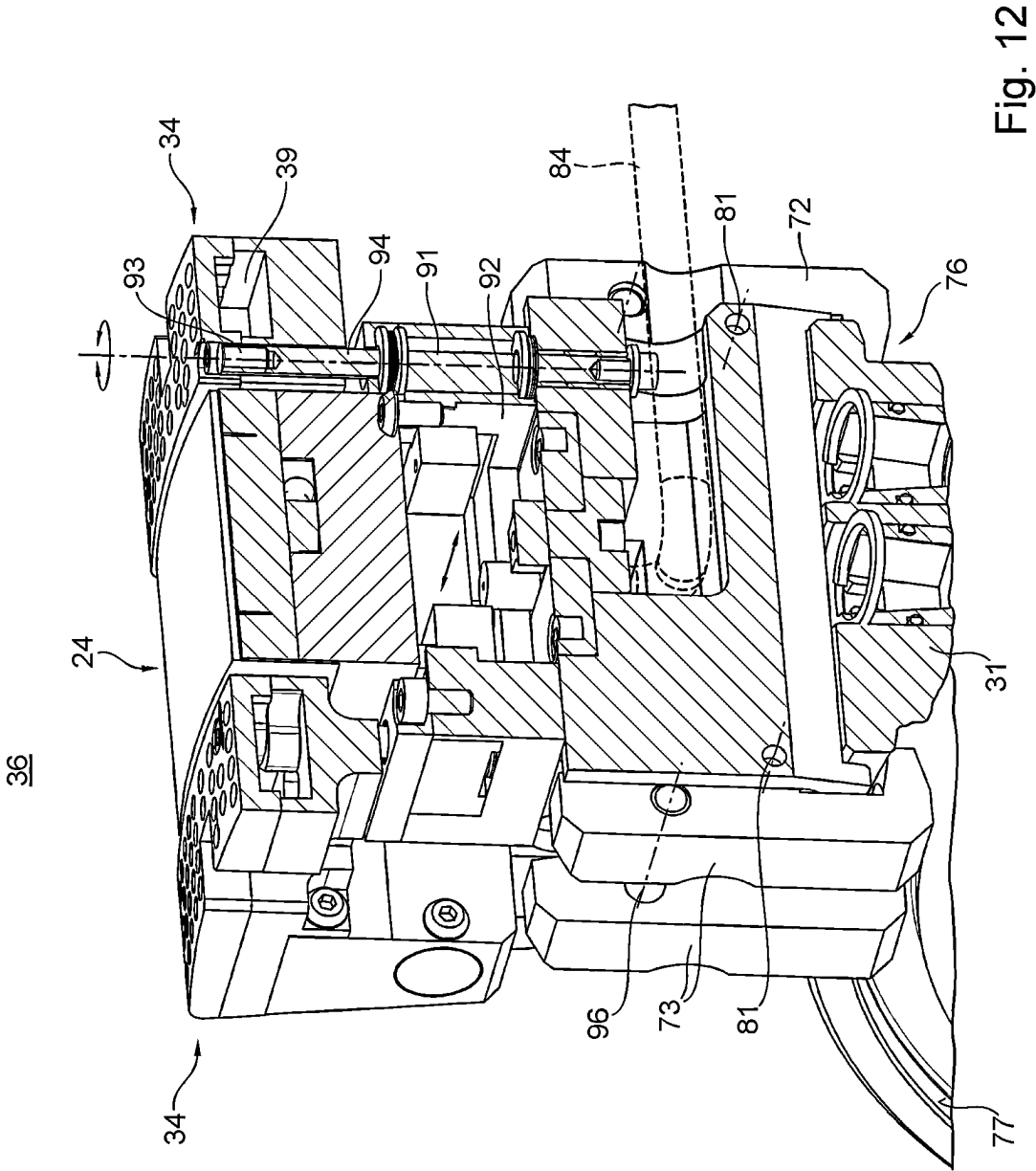


Fig. 12

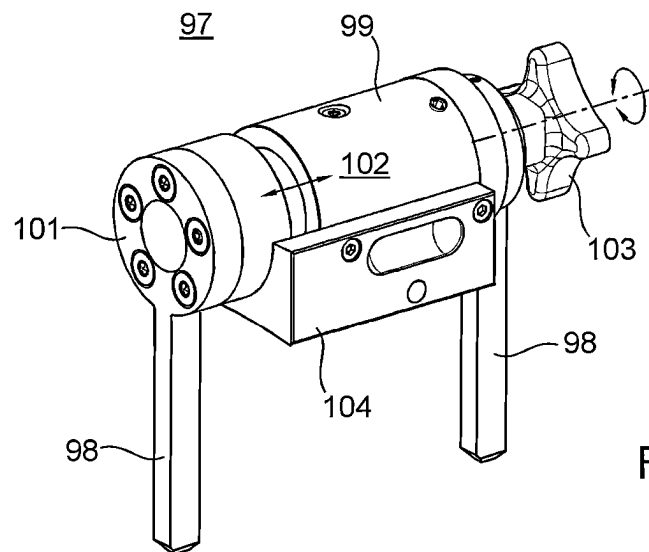


Fig. 13

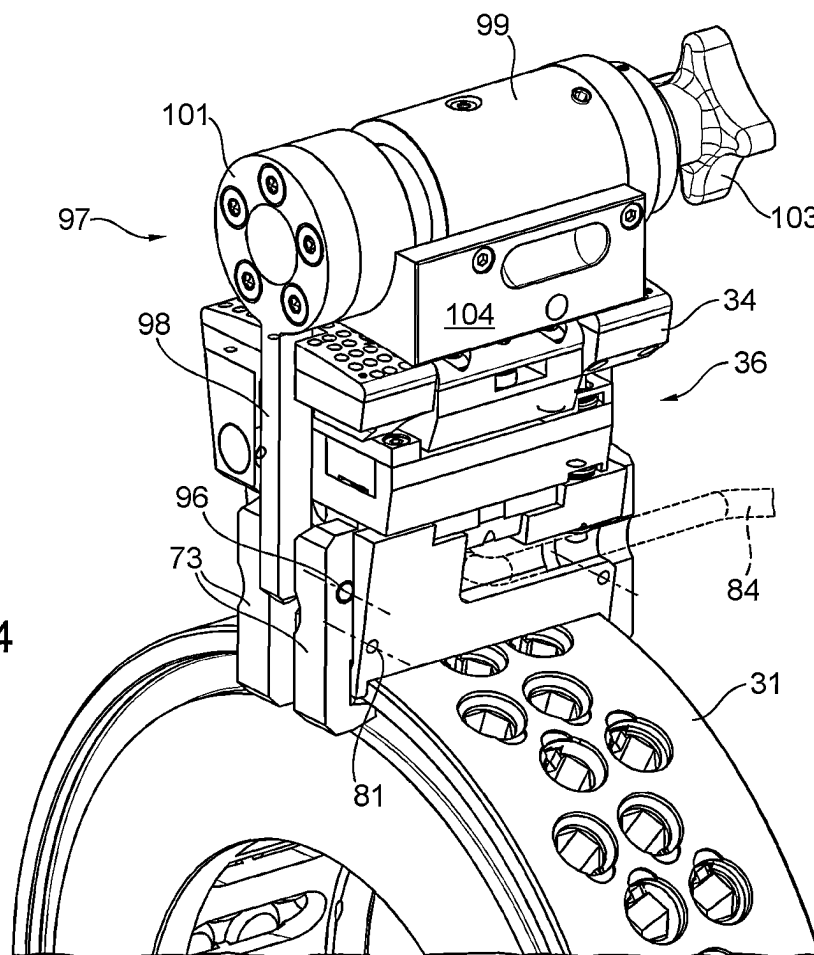


Fig. 14

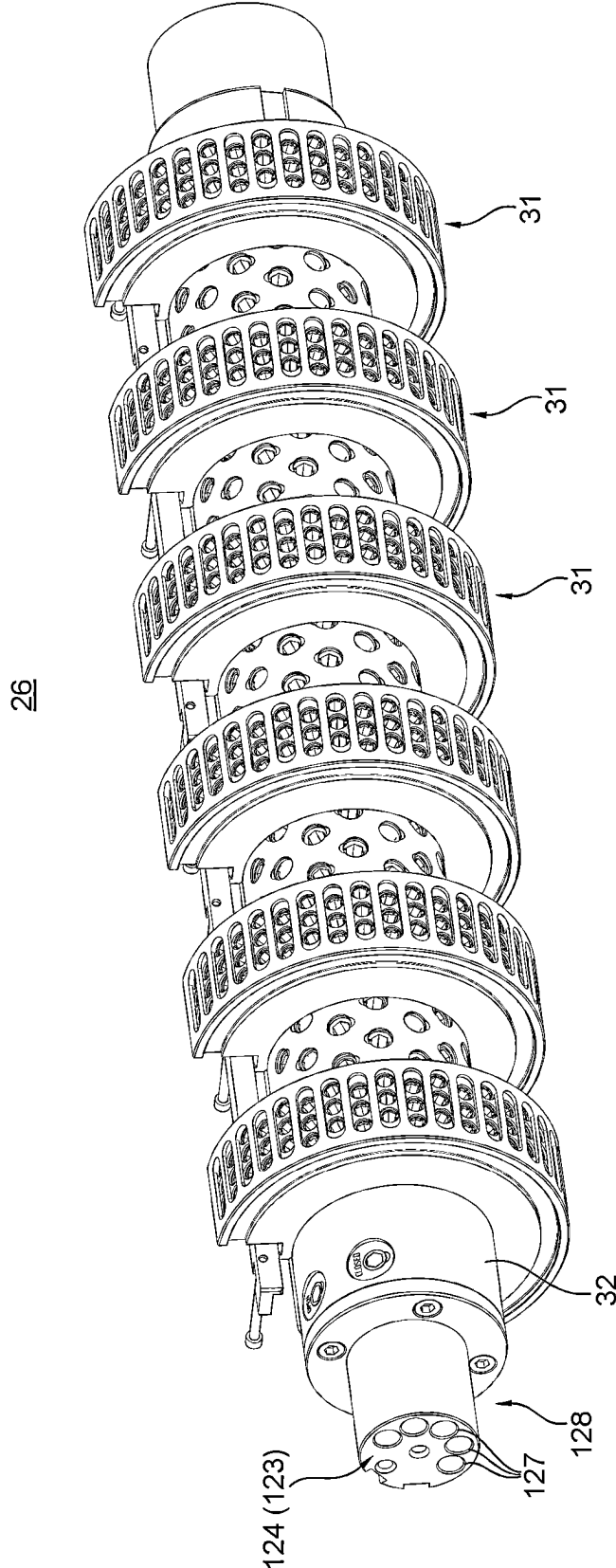


Fig. 15

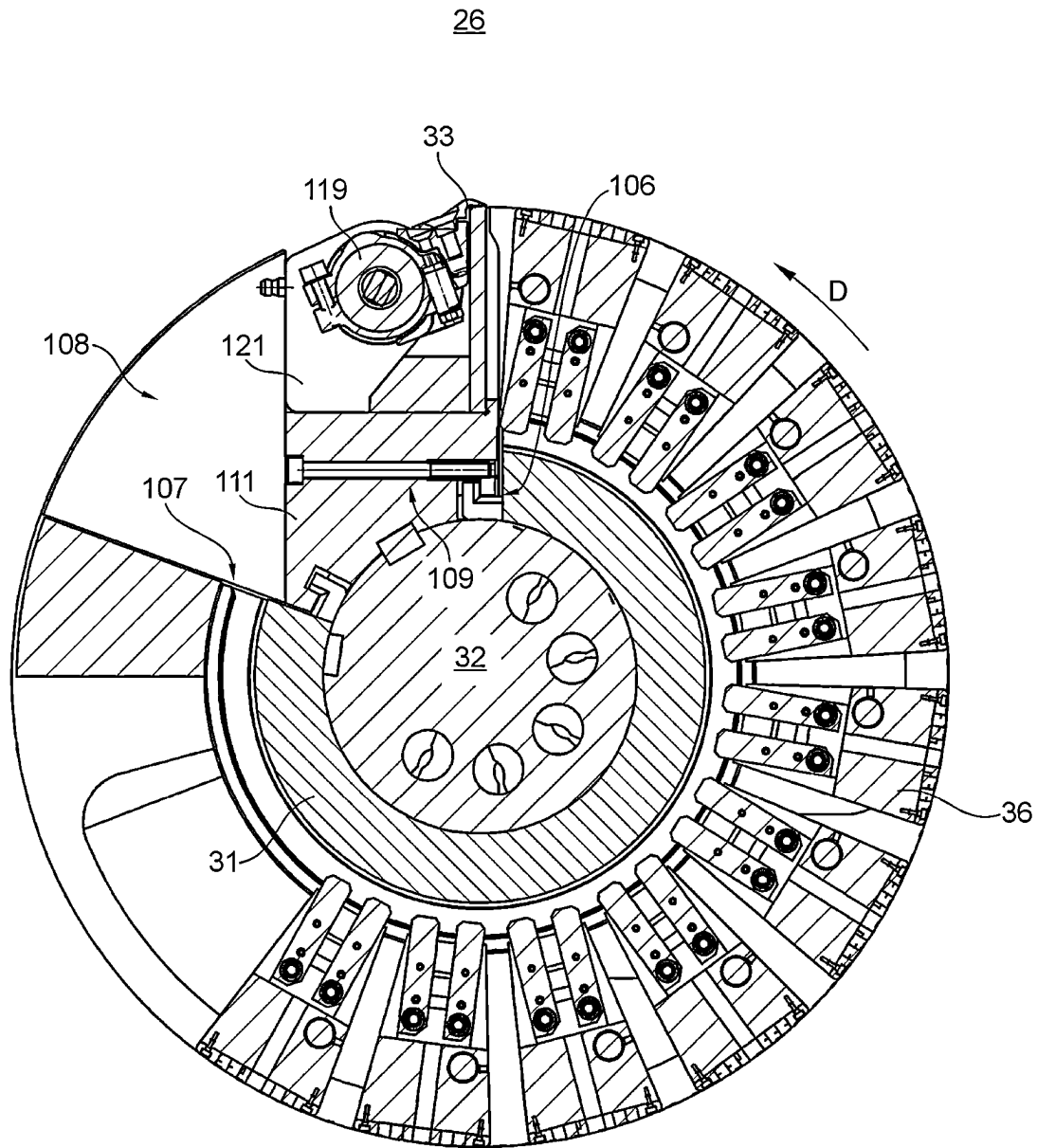


Fig. 16

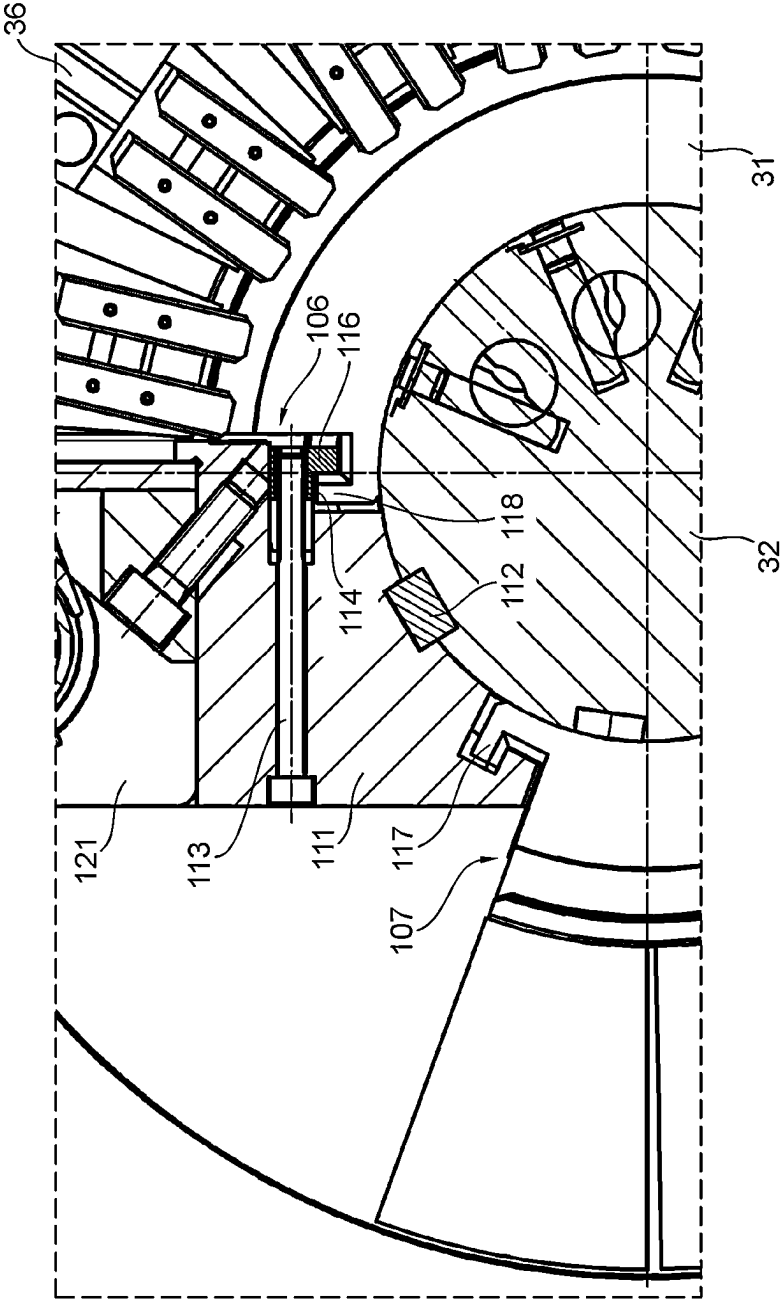


Fig. 17

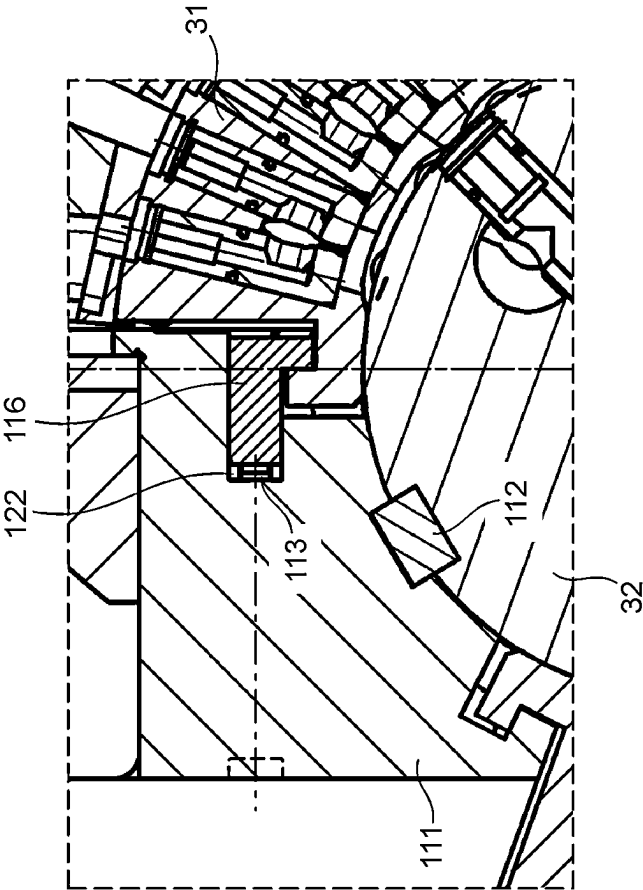


Fig. 18

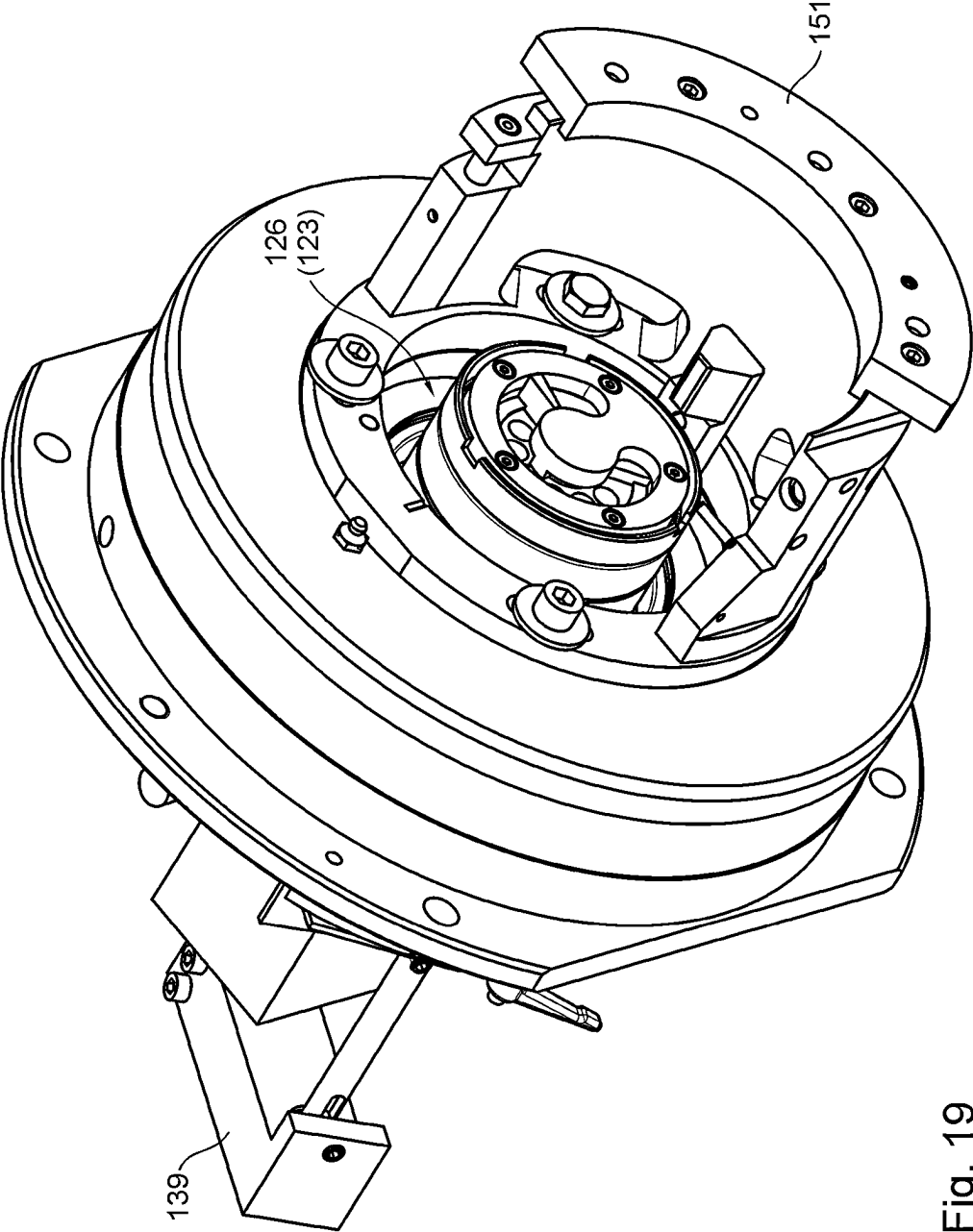


Fig. 19

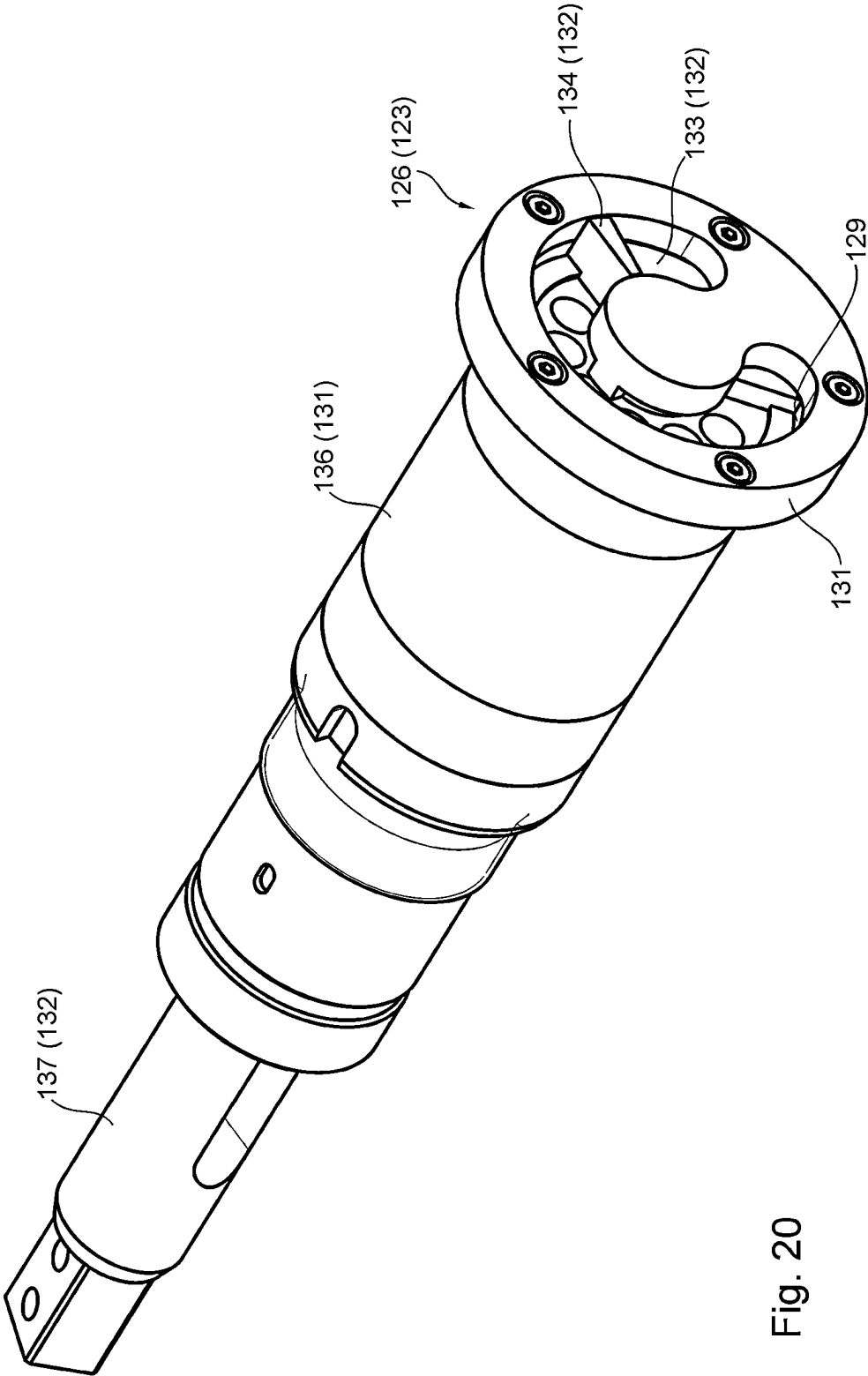


Fig. 20

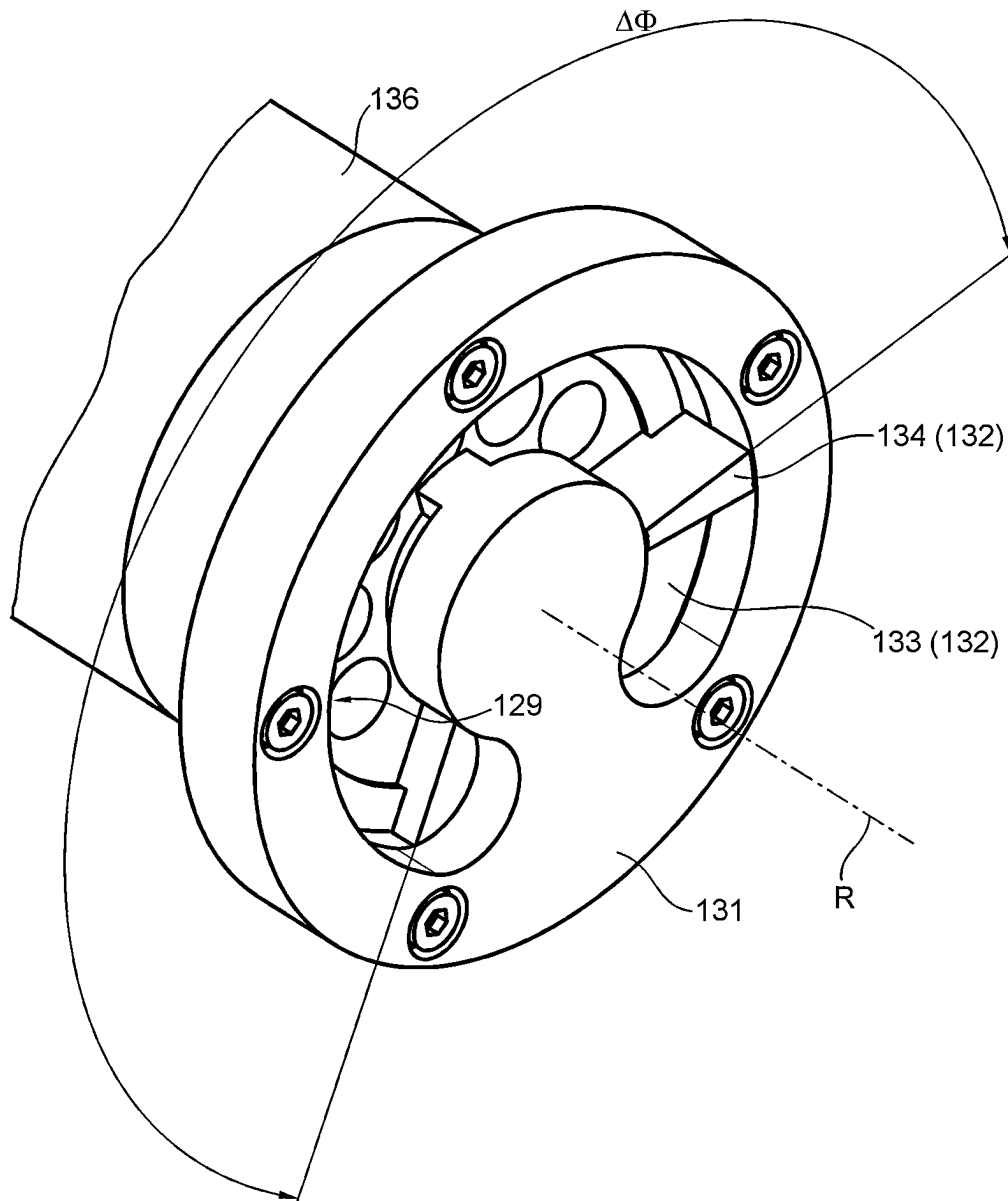


Fig. 21

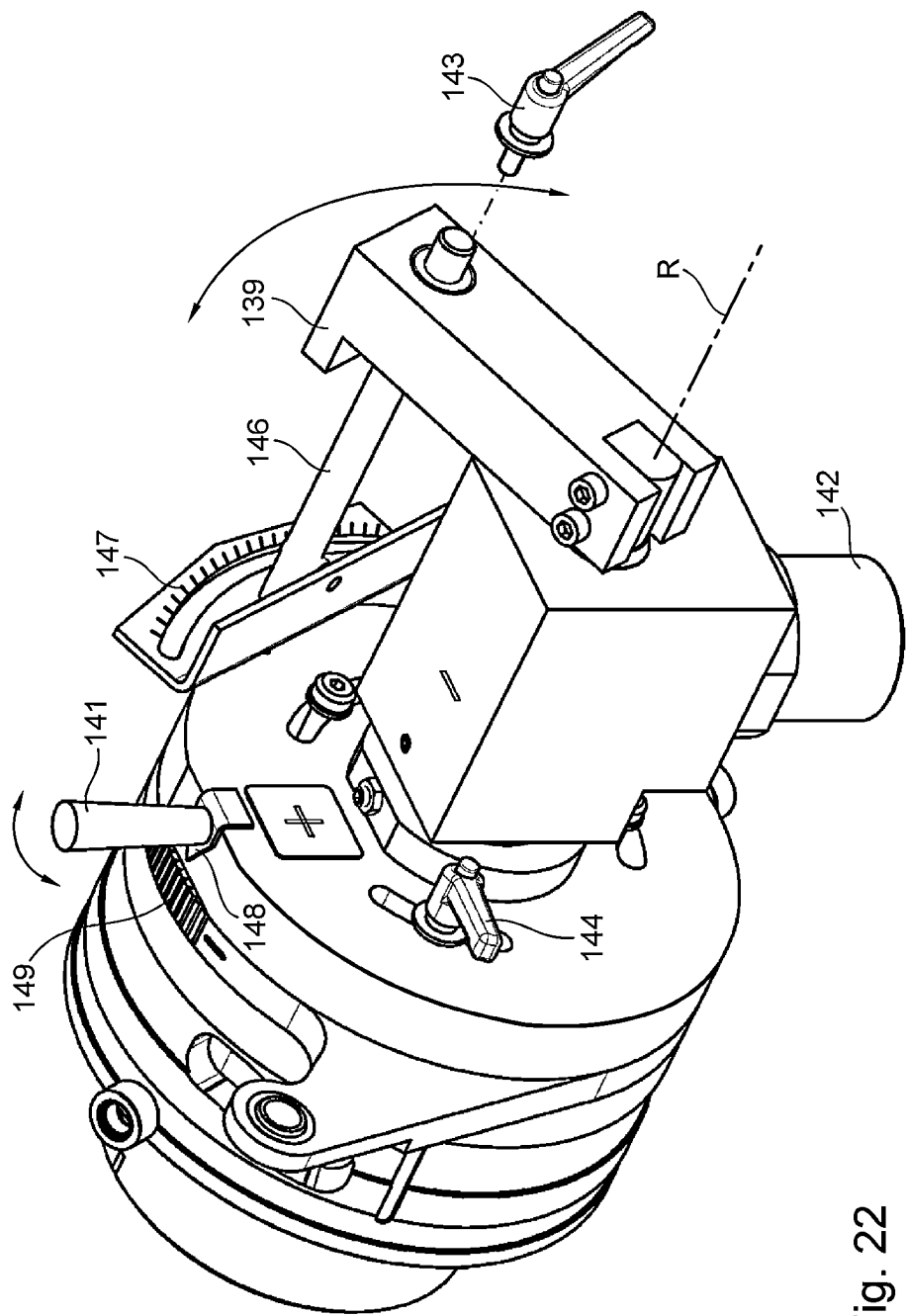


Fig. 22

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**CYLINDER FOR A SHEET-TREATING
AND/OR SHEET-PROCESSING MACHINE
WITH SUCTION AIR OPENINGS AND
MACHINE FOR TREATING AND/OR
PROCESSING SHEET-FORMAT SUBSTRATE
COMPRISING SUCH A CYLINDER**

**CROSS-REFERENCES TO RELATED
APPLICATIONS**

This application is the US national phase, under 35 USC § 371, of PCT/EP2023/071447, filed on Aug. 2, 2023, published as WO 2024/083377 A1 on Apr. 25, 2024, and claiming priority to DE 10 2022 127 807.7 filed on Oct. 21, 2022, and all of which are incorporated by reference herein in their entireties.

TECHNICAL FIELD

Some examples herein relate to a cylinder for a sheet-treating and/or sheet-processing machine with suction air openings and to a machine for treating and/or processing sheet-format substrate comprising such a cylinder. For instance, the cylinder includes holding means around the circumference, by which a substrate sheet to be conveyed over the cylinder can be picked up at the leading end thereof and can be or is held during a rotation of the cylinder over a rotation angle range between when the substrate sheet is received and it is transferred downstream. A rotary union, which includes a rotor that rotates along with the cylinder, is non-rotatably arranged at the end face at a shaft end of a shaft that drives the cylinder or at an end-face cylinder journal. A stator that does not rotate during regular operation is also arranged at the end face. A vacuum pressure can be applied via the rotary union to suction openings or groups of suction openings provided around the circumference of the cylinder while an active rotation angle phase of the cylinder about the axis of rotation thereof is being run through.

Additionally, examples include a machine for treating and/or processing a sheet-format substrate and include substrate infeed, at least one printing mechanism, by which substrate guided on a transport path through the machine is and/or can be printed at least on a first side in a matrix-like manner with multiple-ups having a number m of columns and a number n of rows, a product receiving system, by which processed substrate can be combined into bundles, as well as at least one transport cylinder provided in the substrate path between the substrate infeed and the product receiving system.

BACKGROUND

A printing machine comprising a screen printing unit and a device for aligning magnetic or magnetizable particles contained in the printing ink or the varnish is known from DE 10 2018 212 429 B4, wherein the device comprises a cylinder that has, around the circumference, a plurality of magnetic-field-generating elements arranged in multiple axially adjustable ring elements. The ring elements have suction air openings at the level of the cylinder enveloping for holding conveyed sheets and are connected, or can be connected, to a vacuum line or source via a rotary union.

DE 11 2012 006 348 B4 relates to a combination printing machine and discloses, among other things, a magnetic cylinder with suction openings around the circumference of a rotating cylinder outer body, which are supplied with suction air during rotation from an air chamber extending

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internally over an angular region when passing over the angular region. The air chamber is delimited by two dividing strips firmly attached to an inner shaft and extending in the axial direction.

DE 10 2012 220 401 B4 discloses a transfer drum used to transport a printed sheet, which has an inner tube and an outer tube coaxially to the inner tube, arranged rotatably thereon, which is provided with suction openings around the circumference. An intermediate tube, which is adjustable relative to the inner tube by a limited angle of rotation, is arranged coaxially to the rotationally fixed inner tube between the inner tube and the rotatable outer tube, wherein the intermediate tube, with its inner wall, encloses the circumferential surface of the inner tube. The intermediate tube has at least one groove in the circumferential direction, which extends along only a part of the circumference of the inner tube or the intermediate tube and changes its position in the circumferential direction and thereby the position of the angular range for the passage of the suction air by rotation of the intermediate tube.

DE 1 917 795 A relates to a transport device for sheet-format objects which operates with suction air, wherein a roller casing provided with radial channels is rotatably mounted in the manner of a rotary valve on an inner stator, which has, in its interior, a suction line permanently connected to a suction air source.

DE 10 2014 001 969 B4 relates to a device for changing the format of a sheet transport drum which has an outer sleeve with pneumatic nozzle channels and an inner sleeve with a cover segment for covering the nozzle channels depending on the format.

SUMMARY

It is an object of examples herein to provide a cylinder for a sheet-treating and/or sheet-processing machine with suction air openings and a machine for treating and/or processing sheet-format substrate comprising such a cylinder.

The object is achieved in some examples by the cylinder discussed above, in which the rotary union can be set in terms of a size of a passage angle sector determining the size of the active rotation angle phase and/or in terms of a position of the passage angle sector determining the position of the active rotation angle phase. Additionally, in the case of the machine for treating and/or processing sheet-format substrate, the machine may include the cylinder with these features.

The advantages that can be achieved with the invention consist, for example, in the fact that a sheet can be transported on the cylinder particularly precisely and/or securely, in particular without it being able to perform uncontrolled movements. This is achieved by suction being applied to the sheet to draw it against the cylinder over at least one rotation angle phase.

It is particularly advantageous that a sheet transport can be optimized, especially with regard to a takeover and/or transfer from or to other sheet conveying means. This can be achieved in particular in that the rotation angle phase can be varied with activated suction openings, i.e., with vacuum pressure applied, at least within certain limits in terms of its position and/or size around the axis of rotation of the cylinder.

The rotary union, which is preferably to be arranged at the end face of a shaft driving the cylinder or on an end-face cylinder journal in a rotationally fixed manner, makes complex cylinder installations and/or adjusting mechanisms unnecessary and/or can be retrofitted to existing cylinders.

A cylinder according to the invention for a or of a sheet-treating and/or sheet-processing machine comprises holding means around the circumference, by means of which a substrate sheet to be conveyed over the cylinder is or can be picked up at the leading end thereof and held during a rotation of the cylinder over a rotation angle range between when the substrate sheet is received and it is transferred downstream, and a rotary union, via which, in particular when a vacuum pressure is present on the inlet side and/or a suction air source is connected, vacuum pressure can be applied to suction openings provided around the circumference of the cylinder during an active and/or defined rotation angle phase of the cylinder, wherein the rotary union is adjustable at least with respect to a size of an angle sector determining the size of the defined and/or active rotation angle phase for the passage of the suction air, in short passage angle sector, and/or with respect to a position of a passage angle sector determining the position of the defined and/or active rotation angle phase.

The above-mentioned solution with a rotation angle phase that can be varied in position and/or size or a passage angle sector that can be varied in position and/or size is advantageous with regard to high accuracy and/or good adjustability in the treatment and/or processing of substrate, in particular in the production of optically variable image elements in conjunction with a clamping device that fixes the ring elements, which is also described below, and/or with a design of magnetic and suction elements as modular units, which is also described below, and/or with a movability of individual or all magnetic elements in the axial and/or circumferential direction, which is also described below, and/or with a clamping device that clamps the magnetic elements or modular units, which is also described below.

In a preferred embodiment, the cylinder is designed as a magnetic cylinder and comprises, in the region of its outer circumference, in a matrix-like manner, a number, for example $n \times m$ (with $n, m \geq 0$) of magnetic elements arranged in axially parallel extending rows and in columns extending in the circumferential direction, as well as suction elements with suction openings pointing outwards. In an advantageous refinement, multiple or all of the magnetic elements of columns extending in the circumferential direction are arranged at or on a respective support element, which is open or closed in a ring-like manner and accommodated on a cylinder shaft, the ring-like support element comprising multiple chambers one behind the other viewed in the circumferential direction, which each, independently of one another, have a line connection via corresponding line paths to the rotor of the rotary union on the one hand and to at least one group of suction openings joining at the circumference on the other hand.

A machine according to the invention for treating and/or processing in particular sheet-format substrate comprises a substrate infeed, in the case of possibly web-format starting substrate, for example a cross cutter, at least one printing mechanism, by means of which substrate guided on a transport path through the machine is and/or can be printed at least on a first side in a matrix-like manner with multiple-ups of a number m of columns and a number n of rows, a product receiving system, by means of which processed substrate can be combined in bundles, and at least one transport cylinder provided in the substrate path between the substrate infeed and the product receiving system in the substrate path, which transport cylinder is designed in a manner mentioned above with suction openings and a rotary union connected to these in a line connection.

Further details and variant embodiments can be found in the exemplary embodiments.

BRIEF DESCRIPTION OF THE DRAWINGS

Exemplary embodiments of the invention are illustrated in the drawings and will be described in greater detail below. The figures show:

FIG. 1 an exemplary embodiment of a machine for generating optically variable image elements on a substrate;

FIG. 2 a schematic representation of a substrate printed in print elements with optically variable coating agent, wherein a) shows a state with not yet oriented magnetic or magnetizable particles, and b) a state after alignment of an image-producing part, here by way of example in the form of a digit "1";

FIG. 3 a schematic representation of a printing- and downstream alignment process with an image-producing printing mechanism cylinder and a cylinder comprising magnetic elements, shown by way of example with a substrate sheet widening trapezoidally towards the trailing end;

FIG. 4 a perspective view of an exemplary embodiment of a magnetic cylinder;

FIG. 5 a single representation of a carrier element fitted, by way of example, with multiple magnetic elements one behind the other in the circumferential direction;

FIG. 6 a sectional view through a carrier element fitted with a magnetic element, but in a narrower design compared to FIG. 5;

FIG. 7 a sectional view of a longitudinal section through a cylinder according to FIG. 4;

FIG. 8 an exemplary embodiment of a valve that selectively opens and closes a suction air opening;

FIG. 9 an exemplary embodiment of a less complex design of an action unit comprising a magnetic element and a suction element;

FIG. 10 a top view of an action unit according to FIG. 9, but without magnet and housing;

FIG. 11 a sectional view of an exemplary embodiment of an action unit comprising a magnetic element and a suction element with a recognizable adjusting mechanism for axial positioning of the magnetic element;

FIG. 12 a sectional view of an exemplary embodiment of an action unit comprising a magnetic element and a suction element with a recognizable adjusting mechanism for positioning the magnetic element in the circumferential direction;

FIG. 13 a perspective view of an assembly aid for attaching or removing and/or positioning an action unit;

FIG. 14 a perspective representation of an action unit arranged on a ring element with the mounting aid from FIG. 13 attached;

FIG. 15 a perspective representation of a cylinder inner body fitted with six ring elements;

FIG. 16 a cross-sectional representation of a cylinder with a cylinder inner body, a ring segment-like carrier element arranged thereon and, by way of example, ten action units arranged on the latter;

FIG. 17 an enlarged section of the fastening device for the carrier element;

FIG. 18 a section showing a detail of the fastening device for the carrier element;

FIG. 19 a perspective representation of an end-face connection for a cylinder with a stator of a rotary union that, during normal operation, does not rotate;

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FIG. 20 the part of the rotary union according to FIG. 19 that, during normal operation, does not rotate, with the shafts extending outwards;

FIG. 21 an enlarged section of the two-part stator of the rotary union from FIG. 20;

FIG. 22 a perspective side view of a connection comprising the rotary union for a cylinder with adjusting means for positioning the stator.

DETAILED DESCRIPTION

A machine 01, in particular a securities machine 01, for example a printing machine 01, in particular a securities printing machine 01, treating and/or processing web-format or in particular sheet-format substrate 02, preferably for generating optically variable image elements 03 on a substrate 02, for example an in particular sheet-format printing substrate 02, comprises, for example, an application device 04, for example a printing mechanism 04, by which an optically variable coating agent 06, for example optically variable printing ink 06 or varnish 06, can be applied at at least one application point, for example printing nip 11, to at least a first side of the substrate 02, for example of the printing substrate 02, across the entire surface area or in partial regions in the form of print image elements 08, and, in the case of a machine 01 for generating optically variable image elements 03, a device 07 for aligning particles P that are contained in the optically variable coating agent 06 applied to the substrate 02 and that are responsible for the optical variability (see, for example, FIG. 1). This device 07 is also referred to as an alignment device 07 for short or, since it produces an image of the optically variable pattern or motif as a result of a defined alignment of the particles P, is also referred to as an image-producing alignment device 07 in the following. An application of coating agent 06 that contains particles P onto the printing substrate 02 and an image element 03 obtained by a subsequent image-producing alignment of previously randomly oriented particles P are schematically shown, for example, in FIG. 2 based on an illustration of the digit "1." Here, a) represents a state in which the coating agent 06 has been applied and, for example, is still present in randomly oriented form, and b) represents a state in which an image-producing alignment has taken place.

The print image elements 08 made up of variable coating agent 06, which are applied onto the substrate 02 by the application device 04 prior to the treatment by the alignment device 07, can correspond to the optically variable image elements 03 to be generated in terms of size and position, or possibly may also be larger than these, and possibly can even extend across the surface area of several multiple-ups 09. In the case of larger print image elements 08, for example, an optically variable image element 03 is not generated by alignment on the entire surface area that is coated with optically variable coating agent 06.

The particles P responsible for the optical variability contained here in the coating agent 06, for example the printing ink 06 or the varnish 06, are magnetic or magnetizable, non-spherical particles P, for example pigment particles P, hereafter also referred to as magnetic flakes for short.

The machine 01 is preferably designed to produce multiple-ups 09, for example securities 09, and in particular bank notes 09. This shall in particular also cover the production of intermediate securities products, for example the production of printing substrate 02, in particular in the form of web-format or sheet-format printing substrate sec-

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tions 02, in particular printing substrate sheets 02, using print images of multiple securities 09. The substrate 02 can be formed by, for example, cellulose-based or preferably cotton fiber-based, or at least cellulose-containing or preferably cotton fiber-containing, paper, by plastic polymer or by a hybrid product thereof. It may be present uncoated prior to being coated in the above-described application device 04, or may already have been coated, or it may be unprinted or already have been printed once or multiple times in one or more upstream processes, or may have been mechanically processed in another manner. Preferably, several multiple-ups 09, for example bank notes 09 to be produced or their print images, are arranged on a printing substrate section 02 that is formed by a longitudinal section of web-format substrates 02 or formed by a sheet of a sheet-format substrate 02 in a matrix-like manner, next to one another in rows extending transversely to the transport direction T and one behind the other in columns extending in the transport direction T, or are to be arranged during the course of the processing operation of the substrate 02 (indicated, for example, in FIG. 2 and in FIG. 3).

The machine 01 designed as a printing machine 01 can generally comprise one or more printing mechanisms 04 of arbitrary printing methods. For the sake of simplicity, however, in the embodiment illustrated here it comprises a printing mechanism 04, in particular a printing mechanism 04 operating according to the flexographic printing method, or preferably according to the screen printing method, by which the optically variable coating agent 06 is or can be applied onto a first side of the printing substrate 02. A greater film thickness, compared to other printing methods, can be applied, for example, by the described printing methods, in particular the screen printing method. The expression of the "first side" of the substrate 02 or printing substrate 02 is selected arbitrarily and is intended here to denote the side of the printing substrate 02 onto which optically variable coating agent 06 to be treated downstream by the alignment device 07 is or was or can be applied.

In the illustrated and preferred embodiment, the printing machine 01 comprises a substrate infeed 13, preferably designed as a sheet feeder 13, from which the substrate 02 designed, for example, as a sheet-format printing substrate 02 is or can be fed, possibly via further printing or processing units, to the at least one printing mechanism 04, for example flexographic or preferably screen printing mechanism 04, applying the optically variable coating agent 06, which forms a printing nip 11 for printing, for example, a first side of the printing substrate 02 (see, for example, FIG. 2) between a printing mechanism cylinder 14, in particular a forme cylinder 14, for example a screen printing cylinder 14, and a shared cylinder 17, for example an impression cylinder 17.

Preferably, the printing mechanism 04 comprises a forme cylinder 14 as the image-producing cylinder, including a multiplicity of, in particular like and/or identical, image-producing print elements 18, hereafter also referred to as print motifs 18 or, in particular like and/or identical, groups of image-producing print elements 18 or print motifs 18 around the circumference, which, on a circumferential length corresponding to the print image length, are arranged in multiple, for example a number, for example, of four to eight, in particular five to seven, for example six, columns that are spaced apart from one another transversely to the transport direction T and, on a cylinder width corresponding to the print image width, in multiple rows that are spaced apart from one another in the transport direction T. In the case of a printing mechanism 04 operating according to the

flexographic printing method, these print motifs **18** are designed in the manner of letterpress print reliefs, and in the preferred case of a printing mechanism **04** operating according to the screen printing method, they are designed in the manner of screen printing stencils.

The printing substrate **02** can be fed from the printing mechanism **04** applying the optically variable coating agent **06** to the alignment device **07** via conveying means, for example one or more conveying devices **12** designed as transport cylinders **12**. In the case of a web-format printing substrate **02**, the conveying means could be formed by one or more positively driven and/or non-driven rollers.

After passing through the alignment device **07**, which is described in detail below, the printing substrate **02** can be fed to a further, for example second, conveying device **21** directly or via further conveying means, for example further transport cylinders, and can be fed thereby to a product receiving system **22** for receiving the printing substrate **02** treated and/or processed in the machine **01**, and in the case of sheet-format printing substrate **02** can be fed to a pile delivery **22**. For the preferred case of sheet-format printing substrate **02**, sheet-conveying means, for example one or more transfer cylinders or drums, or, as illustrated here, a conveying device **21** configured, for example, as a revolving gripper conveyor **21**, in particular a so-called chain gripper system **21**, are provided as conveying means, which receive the printing substrate sheets **02** from the transport path section of the alignment device **07** via possibly one or more further transport cylinders and, for example, feed these to the pile delivery **22**.

At least one drying device comprising one or more dryers **23**, for example radiation dryers **23**, directed at the first side of the printing substrate **02**, and possibly a cooling unit (not shown), for example a cooling roller, can be provided at the transport path leading away from the alignment device **07**. In a refinement, an inspection device, for example a sensor device **153**, for example a camera **153**, in particular a line scan camera **153**, cooperating with a cylinder **152**, for example a transport cylinder **152**, in particular an inspection cylinder **152**, can be provided on the transport path between the alignment device **07** and the pile delivery **22**.

In an advantageous refinement, the printing mechanism **04** and the alignment device **07** can be structurally combined, for example in the manner of a module, to form a device **16** for generating optically variable image elements. In a refinement, such a module can, for example, be provided several times one behind the other in the machine **01**. In the advantageous configuration in the manner of a module, the device **16** is or can be inserted into the transport path of the machine **01** to be fitted therewith using input-side and output-side interfaces to corresponding interfaces of a conveyor system, which continues upstream and downstream.

Even though the alignment device **07** described hereafter in detail is essentially arbitrary in terms of the designs, variant embodiments, or configurations thereof, it is preferably provided or can be provided in an above-described machine **01** or printing machine **01**.

The alignment device **07** for creating optically variable image elements **03**, for example for creating the optically variable effect in the optically variable coating agent **06** applied previously, for example in the form of print image elements **08**, onto the substrate **02**, in particular onto the printing substrate **02**, comprises a defined transport path along which the substrate **02** to be conveyed through the alignment device **07** is fed or can be fed from an entrance area, in which the substrate **02** to be treated and having, on the first side thereof, an optically variable coating agent **06**,

is brought or can be brought into operative connection in a defined manner with an alignment device **26** that comprises elements **24** providing magnetic fields, magnetic elements **24** for short, serving as operative elements **24**, preferably in such a way that the magnetic elements **24** of the alignment device **26**, which serve image-producing orientation purposes, and the printing substrate **02** printed with the printing ink **06** containing the particles **P**, move synchronously with respect to one another, at least on a section of the transport path. The alignment device **26** is configured as a magnetically active cylinder **26** here, magnetic cylinder **26** for short, which around the circumference comprises the arrangement of magnetic elements **24** and via which the printing substrate **02** is guided or conveyed, starting from an entrance area, in the direction of an exit area of the alignment device **07**.

The magnetic elements **24** can be formed directly by one-piece or multi-piece magnets **27** themselves or can preferably comprise one or more magnets **27**, which are arranged, preferably detachably or at a mount **28**, for example on or in a base **28**. Here, in general, magnets **27** shall be understood to mean magnetically active devices that, permanently or switchably, at least toward the side of the transport path, induce a magnetic field, which is sufficiently strong, in particular for aligning particles **P** contained in the coating agent **06** on the substrate **02** being guided over the same, as described here. The magnets **27** can be formed by one or more permanent magnets with or without engraving, by solenoids, or by combinations of one or more permanent magnets and/or one or more solenoids. Regardless of whether a single magnet or a combination of multiple magnets, for example permanent magnets and/or solenoids, is involved, the term magnet **27** hereafter shall also be understood to mean multiple magnets **24** that are assigned to the same magnetic element **27** and, in their entirety, form a magnetic unit, unless explicitly expressed otherwise. The magnetic element **24** shall also be understood to include embodiments comprising multiple one-piece or multi-piece magnets **27** that are encompassed by the magnetic element **24** and spaced apart from one another, as they may be employed, for example, if the same multiple-up **09** is to be acted upon by a respective magnetic field at two different points. Such a magnet **27** or such an arrangement of multiple magnets **27** of the same magnetic element **24** can be accommodated in a housing **38** of the magnetic elements **24**, which, for example, is arranged in or on the mount **28** so as to be detachable from the mount **28**.

Generally, it is also possible for two such magnetic cylinders **26** to be provided in the transport path, which are arranged on the same side, or on different sides, of a substrate **02** to be conveyed along the transport path.

In an advantageous embodiment, a drying and/or curing device **19**, for example a radiation dryer **19**, in particular a UV radiation dryer **19**, UV dryer **19** for short, is assigned to the alignment device **07**, which is preferably configured as a UV LED dryer **19** and/or is directed at a point in the transport path at which the substrate **02** cooperates with the magnetic cylinder **26**.

The magnetic cylinder **26** is arranged in the transport path of the substrate **02** to be conveyed, preferably on the second side thereof, so as to point outwardly with the first side, which is coated in particular upstream inline with optically variable coating agent **06**, while passing the magnetic cylinder **26**, in particular while being transported over the magnetic cylinder **26**.

The magnetic cylinder **26** comprises a one-piece, or preferably a multi-piece, cylinder body **29** at or on which the magnetic elements **24** are, preferably detachably, arranged.

The one-piece, or preferably multi-piece, cylinder body **29** can be or is rotatably mounted in a frame. The term of the cylinder body **29** shall encompass both closed structures, i.e., having a more or less closed outer cylinder surface, and open structures, i.e., scaffolding-like or frame-like structures, such as the example illustrated with regard to FIG. 4.

The magnetic cylinder **26** comprises the plurality of magnetic elements **24** in the region of the side facing the substrate path, for example, in the region of the outer circumference, in particular in the region of an outer cylindrical shell surface of the cylinder body **29**, which are used to orient at least a part of the magnetic or magnetizable particles P of the coating agent **06** applied onto the passing printing substrate **02**.

In particular for the case of a plurality of multiple-ups **09** per substrate section, for example per printing substrate sheet or substrate sheet **02**, which is preferred and described here, viewed in the axial direction, multiple columns or groups, in particular a number m ($m \geq 1$) of columns or groups corresponding to the number of columns on the printing substrate section **02**, each comprising multiple rows, in particular a number n ($n \geq 1$) of axially parallel extending rows corresponding to the number of rows of multiple-ups **09** on the printing substrate section **02** to be treated, or, viewed in the transport direction T of the substrate **02** and/or in the circumferential direction of the cylinder body **29**, magnetic elements **24** arranged in a column or group one behind the other are provided or arranged in a matrix-like manner at the cylinder **26**, that is, a number of $n \times m$, in words n times m, with $n, m \geq 1$, magnetic elements **24** are provided in a matrix-like manner around the outer circumference. They are preferably arranged in such a way that, per column or group, the same number n of magnetic elements **24** is provided around the circumference and arranged in axially parallel extending rows and/or, in particular, so that these, when rolled out on the substrate **02**, correspond to the pattern of the image elements **03** to which magnetic fields are to be applied on the substrate **02**, assuming a correct register between the substrate position in the transport direction T and the cylinder angle position. An arrangement in rows or columns shall also be understood to encompass the corresponding lattice-shaped or matrix-shaped arrangement in the case where these are possibly slightly offset from one another in the axially parallel direction for correction or alignment purposes. The n magnetic elements **24** of the columns or groups arranged one behind the other are then, for example, at least arranged one behind the other in the circumferential direction so as to at least partially overlap, when rolled out, along a circular circumferential line and/or end up in multiple-ups **09** of the same column of a substrate **02** to be treated, even if they are possibly slightly offset from one another for correction or alignment purposes. For the axially parallel arrangement, the same applies accordingly to the slight mutual deviations in the circumferential direction that are present, if applicable.

By guiding the substrate **02** over a magnetic cylinder **26** configured in this way, wherein, for example, the first substrate side points outwardly when transported over the first cylinder **26**, it is possible to cause particles P to be aligned or oriented in the region of the image elements **03** provided on the multiple-ups **09** by means of the magnetic elements **24**, that is, here, for example, through the substrate **02**.

The number m of the columns or groups is, for example, four to eight, in particular five to seven, for example six, and/or the number n of the magnetic elements **24** of a

column or group is, for example, two to twelve, advantageously five to ten. The magnetic cylinder **26** or the cylinder body **29** thereof is preferably configured in such a way that the number m of columns or groups and/or the number n of rows or of magnetic elements **24** arranged one behind the other in a column or group can be varied, for example within the above-described boundaries, so as to adapt these to different requirements.

Preferably, the magnetic elements **24** are arranged or can be arranged detachably at the cylinder **26**, preferably in or at a corresponding mount **28** together with the same, in such a way that they, in the mounted state, can be arranged at a defined location around the circumference of the cylinder **26** and can preferably be completely removed from the cylinder **26** and/or can be positioned around the circumference of the cylinder **26** in the axial and/or circumferential directions.

For an above-described matrix-like arrangement, magnetic elements **24** can be arranged and mounted at or in a cylinder body **29** so as to be mounted at the one-piece or multi-piece cylinder body **29** variably in their axial position relative thereto, at least relative to other magnetic elements **24** of the same column or group of magnetic elements **24**. This can be implemented, for example, via axially extending guides around the circumference of the cylinder body **29**, in or on which the relevant magnetic elements **24** are directly or indirectly positioned and can be moved into different axial positions. Such guides could generally be provided individually for individual magnetic elements **24** of a row (see, for example, the embodiment according to FIG. 11), but, if applicable, also continuously for multiple or all magnetic elements **24** of a same row. In this case, the guides could be provided on above-described axially extending carrier elements, which carry all magnetic elements **24** of the same row.

In an advantageous embodiment, the magnetic elements **24** of a respective row or preferably of a respective column, if applicable in addition to an independent axial and/or circumferential positionability of individual or all magnetic elements **24** of the row or column as described in more detail below, can be varied as a whole and independently of an adjacent row or column with respect to their position in the circumferential direction in the case of the row, and as a group with respect to their axial position on the magnetic cylinder **26** or on the cylinder body **29** in the case of the column described here. In the case of a row combined as a group, which is not shown here, in particular multiple, preferably the magnetic elements **24** of all rows, are each combined row by row, for example on axially extending carrier elements, as groups that can be positioned together in the circumferential direction. In the preferred case of columns combined into groups, in particular multiple, advantageously at least the two columns of at least three columns that are closest to the end face, advantageously all columns, are positioned as groups so as to be axially movable in this way in or at the magnetic element carrier **29**, in particular cylinder body **29**.

In the preferred case of columns that are combined into groups, the magnetic elements **24** can be arranged or arrangeable, directly or indirectly, in or at multiple, for example a number m of, for example, four to eight, in particular five to seven, for example six, preferably ring-like carrier elements **31**, for example ring elements **31** here, which are axially spaced apart from one another and preferably an above-described part of which, or preferably all of which, can be positioned in the axial direction on a cylinder inner body **32**, in particular an axially extending cylinder shaft **32**, shaft **32** for short, wherein in turn in each case

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multiple, for example two to twelve, advantageously five to ten, magnetic elements **24** are arranged or can be arranged one behind the other in the circumferential direction in or at these ring elements **31** and at least some of which, or all of which, are arranged or can be arranged so as to be position-
4 able in the circumferential direction (see for example, FIG. 4 and FIG. 5).

For the case of a web-format substrate **02**, the magnetic cylinder **26** can be designed without any holding means acting on the substrate **02** and, for example, with ring elements **31** that are closed in the circumferential direction. For the case of sheet-format substrate **02** preferred here, holding means **33**, for example grippers **33** of a so-called gripper bar, are provided around the circumference of the cylinder **26**, by which a substrate sheet **02** to be conveyed over the cylinder **26** can be picked up at the leading end thereof, and can be held or is held during a rotation of the cylinder **26** over a rotation angle range. A magnetic cylinder **26** configured in this way at the same time serves to transport the substrate **02**. For example, the ring elements **31** are, for example, as is apparent in FIG. 4 and FIG. 5, interrupted in the circumferential direction to receive the holding means **33**. Unless an explicit distinction is made, the term "ring-like" carrier elements or "ring elements" here shall also include non-closed, that is, ring segment-like, elements. In FIG. 5, any fastening means used for fastening, as shown, for example, in connection with an embodiment example according to FIG. 15 to FIG. 18 and also applicable to the embodiments from FIG. 1 to FIG. 14, are not shown further.

In a particularly advantageous embodiment of the magnetic cylinder **26**, individual modular units **36**, which are positioned or can be positioned on or in the cylinder body **29** in a matrix-like manner in columns and rows in the above sense, are provided for multiple or all magnetic elements **24**, hereafter also referred to as action units **36**, in particular magnetic unit **36**, which each comprise both at least one magnetic element **24** and at least one suction element **34**.

In a particularly preferable embodiment of the device **07** for aligning magnetic or magnetizable particles **P**, in several, preferably in all of the *m* columns of magnetic elements **24**, in each case multiple, in particular all of the magnetic elements **24** arranged one behind the other with at least one associated suction element **34** are combined in respective modular units **36** as action units **36** and as such can be positioned as a whole and in each case independently of all other such action units **36** in the circumferential direction and/or can be detached from the cylinder **26**.

The action units **36** each comprise a magnetic element carrier **37**, on or in which the magnetic element **24** is arranged on its outwardly directed side. The at least one suction element **34** can be integrated as part of the magnetic element carrier **37** or arranged as a separate part thereon. Preferably, the action unit **36**, viewed in the axial direction of the cylinder **26**, comprises at least one suction element **34** on each side of the magnetic element **24**. The respective suction element **34** comprises several suction openings **42** in the surface directed outwards, i.e., towards the outside of the cylinder **26** and/or at the level of the cylinder envelope, which are provided, for example, in a cover element **41** covering a suction air channel **39** (see, for example, FIG. 11) in the suction element **34** and preferably detachably fastened above the suction channel **39**. A channel arrangement not apparent in the figures leads from the respective suction air channel **39** through the action unit **36** to a base-side line interface **43**, which is formed, for example, by at least one cut-out **43** (see for example FIG. 11) open towards the inside of the cylinder in a base of the action unit **36** facing the

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inside of the cylinder. Air can be drawn in from the suction openings **42** connected via the channel arrangement and the suction air channel **39** through this at least one cut-out **43** or the line interface **43** formed here thereby and assigned to the action unit **36**.

In an embodiment not shown, the action units **36** can in principle be arranged or arrangeable in a matrix-like manner directly or indirectly on a for example outer cylindrical surface **44** of the axially extending cylinder inner body **32**, in particular the shaft **32**. This cylinder inner body or shaft has, for example, on a longitudinal section which indirectly or directly carries the magnetic elements **24**, radially outwardly directed suction air openings **46** as suction-air guiding line interfaces **46**, which have a line connection, for example via radially extending feedthroughs **47**, for example bore holes **47**, with a channel **48**, for example suction air channel **48**, which extends, for example, axially in the shaft **32** and is to be supplied with suction air from at least one cylinder end.

In the case that the action units **36** are arranged or are to be arranged in a matrix-like manner directly, i.e. directly on the above-mentioned outer cylindrical surface **44** of the axially extending cylinder inner body **32** or the shaft **32**, the action units **36** are or are to be positioned on the outer cylindrical surface **44** in such a way that the line interface **43** on the base of the action unit **36**, here for example the free cross-section of the above-mentioned cut-out **43** in the base of the respective action unit **36**, overlaps with at least one of the line interfaces **43** formed by, for example, suction air openings **46** in the shaft **32**. The aforementioned cut-out **43** forms here, for example, a chamber **43** delimited on the base side by the outer cylindrical surface **44**, wherein a wall in the base area of the action unit **36** completely surrounding the cut-out **43** and an opposite area of the outer cylindrical surface **44** of the shaft **32** form a sealing surface sealing the chamber **43** all around. In this embodiment, the air is drawn in from the suction openings **42** in the relevant suction element **34** via the suction air channel **39** and the channel arrangement, via the line interface **43** of the action unit **36** formed, for example, by the cut-out **43** and at least one suction air opening **46** of the cylinder inner body **32** as well as the suction air channel **48**. For such an embodiment, suitable fastening means, for example in the form of clamping or screw connections, must be provided by which the respective action unit **36** can be fixed to the outer cylindrical surface **44**.

In the particularly advantageous embodiment shown here, however, the action units **36** are not arranged or arrangeable directly on the section of the cylinder inner body **32** or the shaft **32** having the suction air openings **46**, but rather multiple or preferably all of the action units **36** provided for a respective column are arranged as a group at or on an above-mentioned, in particular ring-like carrier element **31**, for example ring element **31**, wherein advantageously at least the outermost carrier elements **31** on both sides, but preferably all of the carrier elements **31** carrying the respective group or column of action units **36**, can be varied in their axial position on the cylinder inner body **32** or the shaft **32**.

The preferably ring-like carrier elements **31** or ring elements **31** each have line interfaces **49**; **51** assigned to the respective carrier element **31** on the side facing inwards, i.e. in the assembled state pointing towards the cylinder inner body **32** or the shaft **32**, and on a side facing outwards, as well as a channel arrangement which connects one or more of the line interfaces **49** on the inside to one or more of the line interfaces **51** on the outside for conducting through suction air. On the inside, for example, cut-outs **49** are

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provided as line interfaces 49 in a wall 52 pointing in the direction of the inside of the cylinder, which each have a line connection via one or more channels 53 extending in the ring element 31 to feedthroughs 54, for example boreholes 54, which lead through the carrier element 31 to line interfaces 51 on the outside and extend radially, for example. Although the openings of individual boreholes 54 could also simultaneously represent the outwardly effective line interfaces 51, preferably one or in particular several of the boreholes 54 also lead on the outside into a cut-out 51 in the outwardly facing wall 56 of the carrier element 31, for example forming the outer line interface 51.

During an application of vacuum pressure to suction openings 42 only while running through a defined rotation angle phase, which is explained in more detail below, the ring element 31 can comprise line connections, separated in segments, between one or more inner line interfaces 49 and one or more outer line interfaces 51, in particular in radially opposite circumferential sections, so that an application of vacuum pressure to one or more inner line interfaces 49 in a segment 45 only requires the presence of vacuum pressure at one or more outer line interfaces 51 assigned to the same segment 45, in particular independently of an application to one or more inner line interfaces 49 of another or adjacent segment 45. For this purpose, direct line connections can be provided between inner and outer line interfaces 49; 51 or, as shown in FIG. 5, for example, chambers 55 assigned to the segments 45, which are separated from one another by walls extending between the outer wall 56 and the inner wall 52 of the ring element 31. Segments 45 or chambers 55 formed in this way can, depending on the desired gradation, extend over a smaller or larger section in the circumferential direction and/or comprise one or more inner and/or outer line interfaces 49; 51 provided one behind the other in the circumferential direction. The application of vacuum pressure or suction air in a certain segment 45 then results in suction through suction openings 42 in a corresponding angular segment Δ_x on the circumference of the cylinder 26.

The ring elements 31 are or will be positioned, for example, on the outer cylindrical surface 44, in particular in such a way that respective or the respective line interfaces 49 on the inside of the ring element 31, here for example the free cross-section of the above-mentioned cut-out 49, overlap with at least one of the suction air openings 46 in the shaft 32 or the cylinder inner body 32. In this case, the above-mentioned cut-out 49 forms, for example, a chamber 49 delimited on the bottom side by the outer cylindrical surface 44, wherein a surface, located outside the cut-out 49, of the inwardly facing wall 52 of the ring element 31 on the inwardly facing side of the ring element 31 forms a sealing surface with an opposite region of the outer cylindrical surface 44 which seals the chamber 49.

Similarly, for example, the action units 36 are or will be positioned in particular on the outward-facing side of the ring element 31 such that the line interface 43 on the base of the action unit 36, here for example the free cross-section of the above-mentioned cut-out 43 in the base of the respective action unit 36, overlaps with at least one of the outer line interfaces 51 on the outward-facing side of the ring element 31. In this case, the cut-out 43 forms, for example, a chamber 43 delimited on the bottom side by the outer wall 56, with the wall completely surrounding the cut-out 43 in the base area of the action unit 36 forming a sealing surface with an opposite region of the wall 56 of the carrier element 31 which seals the chamber 43. In this embodiment, the air is drawn in by the suction openings 42 in the relevant suction element 34 via the suction air channel 39 and the channel

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arrangement, for example via the line interfaces 43; 51 formed by the overlapping cut-outs 43 and 51, the channel arrangement of the ring element 31, the line interface 49 formed, for example, by the cut-out 49 on the inner side of the ring, and at least one suction air opening 46 as well as the suction air channel 48 and, for example, via a rotary union 123, from a suction air source located outside of the cylinder 26. A suction air source is to be understood as any type of air pressure sink which, via a corresponding line connection to suction openings 42, produces a lower pressure than the ambient pressure, i.e., a vacuum pressure at the relevant suction openings 42. This can be, for example, a vacuum pump or possibly a container to which vacuum pressure is applied.

The respective pattern of the suction air openings 46 or line interfaces 46 on the cylinder inner body 32 and the position and shape of the cooperating line interface(s) 43 or cut-out(s) 43 in the base area of the action unit 36 in conjunction with the first above-mentioned variant (without carrier element 31) are preferably matched to one another such that continuous positioning of the action unit 36 in the circumferential direction across at least one adjustment range of two suction air openings 46 spaced apart from one another in the circumferential direction on the shaft 32 is made possible in that in the first variant, at least one of the suction air openings 46 or line interfaces 46 is completely covered by the underside of the action unit 36 in each position located in the relevant adjustment range, while at the same time the opening cross-section of the at least one suction air opening 46 or line interface 46 at least partially overlaps with the bottom-side line interface 43 or cut-out 43 of the action unit 36.

The respective pattern of the suction air openings 46 or line interfaces 46 on the cylinder inner body 32 and the position and shape of the cooperating line interface(s) 49 or cut-out(s) 49 on the inside of the carrier element 31 as well as the position and shape of the cooperating line interfaces 51; 43 or cut-out(s) 49 on the outer side of the carrier element 31 on the one hand and in the base area of the action unit 36 on the other hand in conjunction with the second variant (comprising the carrier elements 31) are preferably matched to one another such that continuous positioning of the action unit 36 in the second variant in the circumferential direction is possible over an adjustment range of at least two line interfaces 51 or cut-outs 51 on the outside of the carrier element 31, in that at least one line interface 51 or cut-out 51 on the outside of the carrier element 31 is completely covered by the underside of the action unit 36, while at the same time the opening cross-section of the at least one line interface 51 or cut-out 51 on the outside of the carrier element 31 at least partially overlaps with the bottom-side line interface 43 or cut-out 43 of the action unit 36.

In connection with a variable positioning, in a particularly advantageous refinement, line interfaces 46; 51 are provided at more points in the axial direction of the cylinder inner body 32 and/or in the circumferential direction on the carrier elements 31 than would be necessary for a single specific configuration for normal operation. However, in order to prevent secondary air from being drawn in through these line interfaces 46; 51 not covered by action units 36 or ring elements 31, closing means 57; 58 are provided, by means of which feedthroughs 47; 54 on the cylinder inner body 32 and/or on the outer circumference of the carrier element 31 supplying line interfaces 46; 51 not covered by the action units 36 or the carrier elements 31 can be selectively closed. In the simplest case, this can be a type of plug which is

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inserted into the relevant feedthroughs 47; 54 for closing and removed from them again as required.

Preferably, however, closing means 57; 58, for example in the form of valves 57; 58, are provided in the selectively closable feedthroughs 47 or 54, which are or can be brought into a closed position in feedthroughs 47 or 54 by feed-
throughs 47 or 54 that are not or only partially directly covered by action units 36 or by carrier elements 31, while at least some of the feedthroughs 47 or 54 are or can be brought into a passage position by line interfaces 46; 51 that are completely covered by action units 36 or by carrier elements 31.

A preferable embodiment of such a closing means 57; 58 is designed in the form of a valve 57; 58, which can be brought selectively into a passage position and into a closing position, without requiring removal or insertion. In an advantageous embodiment, the feedthroughs 47; 54, which are designed in particular as boreholes 47; 54, only have a line connection on one side of the clear cross-section with the channels 48 or 53 of the cylinder inner body 32 or of the carrier element 31, which adjoin on the suction side. For example, in a particularly advantageous embodiment, the valve 57; 58 is formed by a sleeve 57; 58, which on one side has a recess 61 in the lateral wall 62 which, in a rotational position representing a passage position, opens the path into the channel 48; 53 adjoining on the suction side in the cylinder inner body 32 or in the carrier element 31, while in another rotational position it interrupts the connection to the relevant channel 47; 54 by the sleeve wall. In an advantageous embodiment, the sleeve-like valve 57; 58 has, for example at least in a section located further to the outside in the assembled state, an actuating interface 63 which can be brought into engagement with a tool 59 and via which the valve 57; 58 can be rotated between the passage position and the closing position by the corresponding tool 59, in particular without it having to be removed. The corresponding tool interface pair 59, 63 used here is, for example, a polygonal wrench 59 and an inner circumferential section 63 in the sleeve 57; 58 in the form of a polygonal socket 63.

In a refinement of the cylinder 26, a support element 66 is provided between each two columns or groups of modular or action units 36, which has a support surface 67; 68 at the level of the cylinder envelope for supporting the substrate 02 conveyed over the cylinder 26. The support surface 67 can be the outwardly directed cylindrical surface 67 of a circular ring-shaped support disk 64 or the outwardly directed surface 68 of a support plate 71 arranged on a support disk 69, for example made of plastic or metal. The term "circular ring-shaped" or "ring-like" shall also include a support disk 69 that is not completely closed in the circumference, i.e., circular ring segment-like.

In a particularly advantageous embodiment for the fastening of magnetic elements 24 on the cylinder 26, wherein multiple or all magnetic elements 24 of a group are mounted at or on a shared, ring-like carrier element 31 and can be positioned on the carrier element 31 in the circumferential direction, the magnetic elements 24 or a magnetic element carrier 37 supporting the magnetic elements 24 each comprise at least one clamping element 72; 73, for example a clamping lever 72; 73, on both sides when viewed in the axial direction, for example a clamping lever 72; 73, whose ends effective for clamping each engage under a stop surface 74; 77 that, in the assembled state, extends in the circumferential direction on the respective end face of the ring-like carrier element 31 and is directed inwards, i.e. with its surface normal pointing into a cylinder interior, and/or counteracts a radial removal of the magnetic element 24 or

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magnetic element carrier 37 by cooperation with the clamping element 72; 73 situated in the clamping position. In a particularly advantageous embodiment, this stop surface 74; 77 can be an inwardly directed surface of a groove 76; 78 extending in the circumferential direction on the end face of the carrier element 31, into which the clamping element 72; 73 engages with its effective, for example claw-like or clamp-like end. Here, the stop surface 74; 77 or groove 76; 78 extending in the circumferential direction shall encompass a stop surface 74; 77 or groove 76; 78 which is preferably continuous over the full circumference or, as shown, the relevant arcuate section, as well as a stop surface 74; 77 or groove 76; 78 which may be interrupted and continues in several arcuate sections. However, the latter can limit the variability of positioning in the circumferential direction. In addition to the surfaces pointing strictly radially inwards, an "inwardly" directed surface is also to be understood here as surfaces inclined in this direction, the surface vector of which is directed into the interior of the cylinder, but preferably as a circumferential surface per end face focused on the same point on the cylinder axis line, thereby providing the clamping element 72; 73 with a stop directed against a radial removal. In an advantageous variant embodiment, in particular to increase the stability of the seat, two clamping elements 72; 73 that are spaced apart from one another in the circumferential direction or one clamping element 72; 73 comprising two claws cooperating with the carrier element 31 and spaced apart from one another are provided on each side.

Even if the clamping element 72; 73 could basically also be designed as a one-armed lever 72; 73, it is preferably designed in the form of a two-armed lever 72; 73 which can be pivoted about an axis 81, for example pivot axis 81, mounted on the magnetic element 24 or its mount 28 or a modular unit 36 comprising the magnetic element 24, the lever arm of which located closer to the center of the cylinder comprises the part which cooperates with the stop surface 74; 77, for example the claw-shaped or clamp-shaped part, and the lever arm of which located further to the outside is used for actuation. In a preferred manner, the clamping element 72; 73 is preloaded in a self-locking manner, for example by a spring element 79, in particular a compression spring 79, acting between the lever 72; 73, in particular the lever arm located further to the outside, and the magnetic element 24 or the mount 28 or the modular unit 36, in such a way that, in the idle position, i.e. without actuation, it is in the clamping position and holds the magnetic element 24 or the mount 38 or the modular unit 36 on the carrier element 31. The described fastening means offers particular advantages together with a mounting aid 97 described in more detail below.

The above-mentioned type of fastening with the above-mentioned fastening means 72; 73, 74, 77 is basically independent of, but advantageous in conjunction with the design of the above-mentioned modular units 36, in particular action units 36, and/or the special type of suction air guidance or supply and/or of an axial mobility of individual magnetic elements 24, which is described in more detail below, and/or of a mobility of individual magnetic elements 24 in the circumferential direction, which is described in more detail below. The clamping elements 72; 73 make it possible to release the connection from the outside without having to remove the relevant magnetic element 24. Due to continuous adjustability, a release can take place just to the extent that the relevant magnetic element 24 can be posi-

tioned in the circumferential direction against any remaining frictional forces, but without, for example, the risk of tilting, slipping, or falling off.

In some of the figures, for example FIGS. 11, 12 and 14, an optionally provided line 84 is shown or indicated which, in the event that the magnet 27 in the magnetic element 24 is designed to be rotatable by a motor, supplies the motor with signals and/or with electrical energy.

As was already described above in connection with FIG. 2 and FIG. 3, the respective column of image-producing print motifs 18 extending in the circumferential direction of the forme cylinder 14 relates to the same column of multiple-ups 09 provided, or to be provided, one behind the other on the substrate 02. Ideally, these multiple-ups 09 are aligned with one another along the transport direction T and have a uniform width. In cases deviating therefrom, for example when, during an upstream process or due to other mechanical or physical loading, a trapezoidal deformation of the substrate 02 possibly printed previously in the pattern of the multiple-ups 09 has taken place, a geometry that has been changed in this way can be countered by an accordingly varied arrangement of the print motifs 18 on the forme cylinder 14. The print motifs 18 of individual columns are then, for example, not strictly aligned with one another in the circumferential direction, but are located, for example, partially on helical lines that are slightly inclined in relation to the circumferential line (shown, for example, in an exaggerated illustration in FIG. 3 for better perception). The width of the multiple-ups 09 on the substrate 02 increases, for example, from the leading to the trailing end of the substrate section or substrate sheet 02 or, for example with appropriate reverse infeed at the entrance of the printing machine 01, possibly in the opposite direction. However, there may also be other reasons for a deviation in the relative position between the axial position of individual magnetic elements 24 and the target position for their effect on the substrate 02, such as, for example, a slightly incorrect axial positioning of the magnetic elements 24 on the cylinder 26, etc.

In principle, irrespective of an arrangement of the magnetic element 24 in an above-mentioned modular unit 36 and/or of the design of an above-mentioned fastening device and/or of adjustability in the circumferential direction, but preferably in conjunction with one or more of the described advantageous embodiments, in a particularly advantageous embodiment at least in multiple, preferably in all of the columns or groups of magnetic elements 24 extending in the circumferential direction, in each case at least one of the magnetic elements 24 is mounted directly or indirectly on the cylinder body 29 of the magnetic cylinder 26 so as to be adjustable or movable at least in the axial direction independently of at least one other magnetic element 24 of the same column or group. Preferably, multiple, advantageously all except one, in particular advantageously, however, all magnetic elements 24 of the same group are mounted so as to be axially movable, independently of other magnetic elements 24 of the group, and/or multiple, advantageously all except one, or all magnetic elements 24 of at least the two groups, in particular all columns or groups, of at least three columns or groups that are closest to the end face are mounted so as to be axially movable in or at the cylinder body 29, independently of other magnetic elements 24 of the particular column or group. This allows the above-mentioned random or systematic relative deviations of individual magnetic elements 24 in the axial position to be readjusted or corrected. In particular in conjunction with the above-mentioned indirect mounting of the magnetic elements 24

via magnetic element carriers 37, which are provided directly or indirectly on the cylinder body 29 via the above-mentioned carrier elements 31, such axially adjustable magnetic elements 24 are preferably axially adjustable at the relevant magnetic element carrier 37 and relative thereto.

In a particularly advantageous embodiment of the cylinder 26 with the $n \times m$ magnetic elements 24 arranged in a matrix-like manner, at least two or all of the magnetic elements 24 provided in the same column one behind the other are mounted at or on an above-mentioned shared carrier element 31 and can be varied together with the same and independently of an adjacent group with regard to their axial position in or on the cylinder 26, wherein in addition the at least two or preferably all magnetic elements 24 of this or preferably each column are arranged on respective magnetic element carriers 37, which can be positioned independently of one another in the circumferential direction on the shared carrier element 31 and/or can be detached from the carrier element 31 and are mounted on the relevant magnetic element carrier 37 so as to be adjustable in the axial direction within an adjustment range, for example, of at least 1 mm in total, preferably at least 2 mm.

In this embodiment, the axially movable magnet 27 or the mount 28 is thus supported indirectly via the associated magnetic element carrier 37, which carries the respective, at least axially movable magnetic element 24 and is preferably itself variably positionable on the ring element 31 in the circumferential direction.

In a simple and less complex embodiment (see for example FIG. 10), the respective magnetic element 24 or the mount 28 is fastened in or on the magnetic element carrier 37 or its mount 28 by means of a fastening means 83, for example a screw 83, in such a way that after at least partial loosening of the fastening, for example by at least partial loosening of the screw 83 by means of an appropriate tool, the magnetic element 24 or the mount 28 is released to such an extent that it can be moved axially, at least within a relevant adjustment range on the magnetic element carrier 37. The fastening means 83, for example in the form of a screw 83, is accessible, for example, through a cut-out in the form of an elongated hole 82 in a base area of the mount 28 receiving the magnetic element 24 after removal of the magnetic element 24.

However, a movement or adjustment of the magnetic element 24 or of the mount 28 encompassed by it in the axial direction is preferably carried out, in contrast, for example, to a purely manual and/or tool-free movement, via mechanical adjusting means 86, 87, 89, in particular comprising a gear.

Although the adjusting means 86, 87, 89 effecting an axial movement can be realized by any suitable mechanisms or gears, in the illustrated and particularly advantageous case these comprise a gear converting, for example directly or indirectly, a rotational movement, in particular on the input side, into a linear movement, in particular of the magnetic element 24 or of the mount 28 carrying the magnetic element 24, in particular an eccentric drive which converts a rotational movement of an eccentric 86, for example formed by an eccentrically mounted shaft section 86, into a linear movement, extending axially here, of a slide 87, for example of a support element 87 directly or indirectly carrying the magnetic element 24 or its mount 28, which is directly or indirectly operatively connected via a contact with the effective surface on the eccentric jacket side and is mounted linearly movably in or on the magnetic element carrier 37. The eccentric 86 preferably extends with its axis of rotation

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radial to the cylinder 26 and/or can be actuated directly or indirectly from the outwardly facing cylinder side. For this purpose, for example, a shaft 89 surrounding the eccentric 86 or continuing outwards has an actuating interface 88, for example a polygonal socket 88, in the region of its outwardly pointing end, which can be actuated, in particular pivoted, by means of a corresponding tool, here for example a polygonal wrench. As an alternative to the eccentric 86 positioned radially with the axis of rotation, a tangential position or a position parallel to the tangent is also conceivable, wherein this eccentric can then be actuated, for example, from a side pointing in the circumferential direction or via an angle gear from the outside.

An adjustment range in the axial direction, as viewed from a center position, is, for example, at least ± 1.0 mm (i.e., a total adjustment travel of at least 2 mm), preferably at least ± 1.2 mm, for example ± 1.5 mm.

In the above embodiment as an action unit 36 comprising at least one suction element 34, in one variant embodiment the at least one suction element 34 can be axially movable together with the magnetic element 24 on the magnetic element carrier 37. A corresponding suction air feed-through, for example via relative movable sealing surfaces or a flexible line, must be provided.

There may also be deviations in the relative position between the position of individual magnetic elements 24 in the circumferential direction of the cylinder 26 and the target position for their action on the substrate 02 in the transport direction T, which may have a wide variety of reasons, such as, for example, limited options for rough and/or manual pre-positioning on the cylinder body 29 or, in particular, on a carrier element 31 that may be provided.

In principle, irrespective of the arrangement of the magnetic element 24 in an above-mentioned modular unit 36 and/or of the design of an above-mentioned fastening device and/or an above-mentioned adjustability in axial direction, but preferably in conjunction with one or more of the aforementioned advantageous embodiments, in a particularly advantageous embodiment, at least in several, preferably in all axially extending rows of magnetic elements 24, at least one of the magnetic elements 24 is mounted directly or indirectly on the cylinder body 29 of the magnetic cylinder 26 so as to be adjustable or movable at least in the circumferential direction independently of at least one further magnetic element 24 of the same row. Preferably, in multiple, in particular all, rows, multiple, advantageously at least all but one, in particular advantageously, however, all magnetic elements 24 of the same row are mounted so as to be axially movable independently of other magnetic elements 24 of the row.

Instead of or in addition, in a particularly advantageous embodiment of the cylinder 26 with the magnetic elements 24 arranged in a matrix-like manner, at least two magnetic elements 24 provided one behind the other in the same column are arranged on or in magnetic element carriers 37 which differ from one another and can be positioned independently of one another in the circumferential direction on the cylinder 26, wherein the at least two, in particular all, magnetic elements 24 arranged on the respective magnetic element carriers 37 are mounted so as to be adjustable relative to the magnetic element carrier 37 carrying the magnetic element 24 in the circumferential direction within an adjustment range, for example of at least 1 mm in total, preferably at least 2 mm. This preferably applies to at least two or all magnetic elements 24 of all columns.

A movement or adjustment of the magnetic element 24 or of the mount 28 encompassed by it in the circumferential

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direction is preferably carried out here, in contrast, for example, to a purely manual and/or tool-free movement, via mechanical adjusting means 91, 92, 94, in particular comprising a gear.

In addition to a movement on a circular arc-like path, an adjustment or adjusting movement in the circumferential direction in the present meaning shall also explicitly include a movement along a linear movement path extending tangentially or parallel to the tangent on the circumference, over the relevant adjustment range. As this is generally a very small relevant adjustment range compared to the cylinder diameter, the linear adjustment path does not generally lead to impermissibly large imaging errors.

Although the adjusting means 91, 92, 94 effecting a movement in the circumferential direction can be realized by any suitable mechanisms or gears, in the illustrated and particularly advantageous case these comprise a gear converting, for example directly or indirectly, a rotational movement, in particular on the input side, into a linear movement, in particular of the magnetic element 24 or of the mount 28 carrying the magnetic element 24, in particular an eccentric drive which converts a rotational movement of an eccentric 91, for example formed by an eccentrically mounted shaft section 91, into a linear movement of a slide 92, for example a support element 92 directly or indirectly carrying the magnetic element 24 or its mount 28, which is directly or indirectly operatively connected via a contact with the effective surface on the eccentric jacket side and is mounted linearly movably in or on the magnetic element carrier 37. In the above sense, the linear movement shall be both a rectilinear movement, which is preferable because of the complexity involved, but also possibly a movement on a circular arc. The eccentric 91 preferably extends with its axis of rotation radially to the cylinder 26 and/or can be actuated from the outward-facing cylinder side. For this purpose, for example, a shaft 94 surrounding the eccentric 91 or continuing outwards has an actuating interface 93, for example a polygonal socket 93, in the region of its outwardly pointing end, which can be actuated, in particular pivoted, by means of a corresponding tool, here for example a polygonal wrench. As an alternative to the eccentric 91 positioned radially with the axis of rotation, a tangential position or a position parallel to the tangent is also conceivable, wherein the eccentric can then be actuated, for example, from a side pointing in the circumferential direction or via an angle gear from the outside.

An adjustment range in the circumferential direction is, for example, at least ± 1.0 mm (i.e., a total adjustment path of at least 2 mm), preferably at least ± 1.2 mm, for example ± 1.5 mm, when viewed from a central position.

In the above embodiment as an action unit 36 comprising at least one suction element 34, in one variant embodiment the at least one suction element 34 can be movable together with the magnetic element 24 on the magnetic element carrier 37 in the circumferential direction. A corresponding suction air feed-through, for example via relative movable sealing surfaces or a flexible line, must be provided.

In the event that both an axial as well as a circumferential adjustability of the magnetic elements 24 on the respective magnetic element carrier 37 in the circumferential direction is provided, the two slides 87; 92 can be arranged directly or indirectly on top of and/or above one another in the manner of a cross guide.

In one of the above-mentioned embodiments, the relevant magnetic element 24, in a refinement, can be adjusted in the axial and/or circumferential direction by a remotely operable

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drive means, for example an electric motor driving the eccentric **86**; **91**, for example via a gear reducer.

Basically independent of an arrangement of the magnetic element **24** in an above-mentioned modular unit **36** and/or of an above-mentioned adjustability in the axial direction and/or of an above-mentioned adjustability in the circumferential direction, but preferably in conjunction with one or more of the advantageous embodiments described above, a mounting aid **97**, as was already mentioned above, is provided, which can be placed on the magnetic element **24** or on a magnetic element carrier **37** carrying the magnetic element **24** or a modular unit **36** comprising the magnetic element **24**, and by means of which the clamping fit or a clamping connection between the clamping elements **72**; **73** on both sides and the carrier element **31** can be released. In a preferred manner, the clamping can be released and opened by the mounting aid **97** or a drive means **102**, in particular a manually operable drive means **102**, comprised by the mounting aid **97** not only in such a way that the magnetic element **24** or the modular unit **36** comprising it can be removed from the supporting carrier **31**, but can also be released in an intermediate position in the strength or the degree of opening of the clamping to such an extent that the magnetic element **24** or the modular unit **36** is not yet completely free, but can be positioned on the carrier element **31** overall in the circumferential direction. A degree of opening can be adjustable in such a way that although there is still contact between the clamping elements **72**; **73**, positioning is possible while overcoming any slight frictional forces that may still exist. For this purpose, the actuating arms **98** are preferably continuously positionable by the drive means **102** over an adjustment path between a clamping position, in which the clamping elements **72**; **73** develop the full clamping force on the carrier element **31**, and a position in which the clamping is released to such an extent that the magnetic element **24** or the magnetic element carrier **37** carrying it can be removed from the carrier element **31**.

In order to be able to accomplish a simple actuation from the outside of the cylinder and/or, in particular, also such a defined opening, the assembly aid **97** comprises, in addition to a base **104** which can be placed on the relevant magnetic element **24** or on the relevant modular unit **36**, actuating arms **98** on both end faces, which extend in the radial direction to both end faces of the magnetic element **24** or of the modular unit **36** and can be brought or are brought into operative connection with the respective end-face clamping element or elements **72**; **73** for their actuation. Furthermore, the assembly aid **97** comprises the above-mentioned drive means **102**, in particular a positioning drive **102**, by means of which the actuating arms **98** can be brought into a first position in which they open the clamping elements **72**; **73**—for example against the above-mentioned spring force, to such an extent that the magnetic element **24** or the modular unit **36** can be attached to the carrier element **31** or completely detached therefrom, up to a second position in which the clamping elements **72**; **73** develop the full clamping force on the carrier element **31** without the actuating arms **98** absorbing any force that is directed against the clamping force. Preferably, all intermediate positions can be adjusted by the drive.

In the above-mentioned design of the clamping elements **72**; **73** as two-armed levers **72**; **73**, each of the actuating arms **98** engages directly or indirectly on the lever arm located further out and can be moved towards each other by the drive means **102** to open the clamping connection, i.e., in each case in the direction of the carrier element **31**, and can be moved apart again to close the clamping connection.

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In the above-mentioned case of two clamping elements **72**; **73** arranged next to each other, these are coupled to each other, for example, via a coupling element **96** connecting the two lever arms located further out, for example, a connecting axis **96** mounted in both outer lever arms, which serves as an engagement point for the respective actuating arm **98**, for example simultaneously. In the case of a single clamping element **72**; **73**, the respective actuating arm **98** can act directly or indirectly on the lever arm of the relevant clamping element **72**; **73** located further out.

In principle, any drive mechanism is conceivable as drive means **102**, by means of which the two opposing actuating arms **98** can be moved towards and away from each other in the above sense. However, a drive mechanism with a self-locking gear, such as is provided, for example, by a screw drive, is preferred here. The drive means **102** thus comprises, for example, a first part **99** carrying the actuating arm **98** on one side, for example a first bushing **99**, and a second drive part **99** carrying the actuating arm **98** on the other side, for example a second bushing **101**, which is mounted so as to be non-rotatable but axially movable relative to the first part **99**, as well as an internally formed screw drive, by means of which the parts carrying the actuating arms **98** can be moved apart and towards each other via a threaded spindle (not shown), which can be rotated for example via a manual actuating interface **103**, such as a rotary handle **103**, on the one hand, and an internal thread on the other of the two parts **99**; **101** of the drive means **102**.

In the design of the magnetic cylinder **26** with carrier elements **31**, which are variable in their axial position, in particular ring-like carrier elements **31**, these axially positionable carrier or ring elements **31** can in principle be fastened in any manner that enables a releasable connection between the respective carrier element **31** and the cylinder inner body **32** and an axial relative movement. In particular, a connection is particularly advantageous in which, in the region of suction-air conducting line interface pairings of line interfaces **46** on the shaft **32** and cooperating line interfaces **49** on the inward-facing wall **52** of the ring element **31**, the surfaces surrounding these line interfaces **46**; **49** are pressed together by the connection in such a way that they form a sealing surface that is substantially closed to prevent the passage of suction air.

Basically independent of an arrangement of the magnetic element **24** in an above-mentioned modular unit **36** and/or of an above-mentioned adjustability in the axial direction and/or of an above-mentioned adjustability in the circumferential direction and/or of an above-mentioned clamping device for clamping the magnetic elements **24** or mounts **28** or modular units **36**, but preferably in conjunction with one or more of the aforementioned advantageous embodiments, a tensioning device is provided for fastening in a preferred embodiment, for the fastening of ring elements **31**, by means of which the carrier or ring element **31** can be clamped onto the cylinder inner body **32**, which is designed in particular as a shaft **32**, in such a way that an aforementioned sealing surface can be formed. It is helpful to design the ring element **31**, which is actually circular ring segment-like here, i.e. not completely closed in the circumference, in such a way that an inner diameter of the ring element **31** in the segment angle range is slightly larger, for example 2 to 50 μm , in particular 5 to 20 μm , than an outer diameter of the cylinder inner body **32**, which is designed as a shaft **32**, in the cooperating circumferential region.

In the case of a cylinder **26** with magnetic elements **24** arranged in columns, as described above, the magnetic elements **24** of multiple or all columns are provided as a

respective group at or on a respective carrier element 31. The respective carrier element 31 is explicitly designed here as a ring segment-like carrier element 31, i.e., interrupted over a circumferential section or intermediate angle, and has a leading and a trailing end 106; 107 with respect to a production direction of rotation D. The production direction of rotation D is defined, for example, by the arrangement of a gripper bar already mentioned above, which has grippers 33 opening and closing during operation at the leading end 106 of the segment-like ring element 31 for receiving a substrate sheet 02. The respective carrier element 31 is releasably arranged on the cylinder inner body 32 encompassed by the cylinder 26 and is variable in its axial position in the released state. In order to now fasten the carrier element 31 in a desired position on the cylinder inner body 32, a tensioning device 108 is or will be provided on the cylinder inner body 32 when the carrier element 31 is mounted on the cylinder inner body 32 in the region between the leading and trailing ends 106; 107 of the ring-segment-like carrier element 31, by means of which the two ends 106; 107, which are spaced apart from one another in the circumferential direction, can be acted upon by a force directed towards one another in the circumferential direction via adjusting means 109 encompassed by the tensioning device 108. As a result, the segment-like ring element 31 is pressed tightly against the circumferential surface of the shaft 32, possibly by a slight elastic deformation, so that a sealing surface as mentioned above is created.

The tensioning device 108 engages in particular at the two ends 106; 107 of the carrier element 31 and can be varied in its length, effective for the engagement on both sides, in the circumferential direction via the adjusting means 109, encompassed by the tensioning device 108.

Preferably, the tensioning device 108 comprises a tensioning strip 111, which is arranged in the region between the leading and trailing ends 106; 107 of the carrier element 31 on the circumference of the cylinder inner body 32 and is secured at least toward one side to prevent a relative movement with respect to the cylinder inner body 32 in the circumferential direction. Preferably, however, the tensioning strip 111 and the cylinder inner body 32 are secured to prevent rotation in the circumferential direction by pairs of stops acting in both directions of rotation. Such securing can be realized, for example, by corresponding deviations of the inner circumferential line of the ring element 31 and the outer circumferential line of the cylinder inner body 32 acting as stop pairs. In an advantageous embodiment shown here, however, such a relative anti-rotation lock device is provided by a so-called fitted element 112, also commonly referred to as a feather key 112, which is anchored, for example, in the outer cylindrical surface of the cylinder inner body 32 and cooperates with a recess, in particular a groove, in the tensioning strip 111 in a fitting manner or vice versa. A fitted element 112 with a correspondingly cooperating recess is advantageous in such a way that simple radial equipping of the cylinder inner body 32 with the tensioning strip 111 is made possible. In addition to the anti-rotation device, fastening means (not shown), for example screws, can be provided by which the tensioning strip 111 can be fastened radially on the cylinder inner body 32.

Preferably, the tensioning strip 111 can then be removed from the cylinder inner body 32 in the relaxed, i.e. force-free, state of the tensioning device 108 immediately or after the above-mentioned fastening means have been loosened with the carrier element 31 still remaining on the cylinder inner body 32 or can be inserted on the cylinder inner body

32 in the region of the interruption when the ring element 31 has already been positioned on the cylinder inner body 32.

In an advantageous embodiment, the tensioning device 108 engages at one of the ends 107; 106, preferably at the trailing end 107, statically, i.e. in a fixed circumferential relative position between the tensioning strip 111 and the relevant end 107; 106, and engages at the other, preferably the leading end 106 via the adjusting means 109 in a distance-variable manner, i.e. in a variable circumferential relative position between the tensioning strip 111 and the relevant other end 106; 107. This means, for example, that with the adjustment, the point of engagement and thus the relevant end 106; 107 can be moved closer to the tensioning strip 111 or, for example by the elastic restoring force in the ring element 3, returned to the initial position.

For static engagement, for example, a positive fit effective in the circumferential direction is provided via a pair of stops effective between the relevant end 107; 106 and the tensioning strip 111. The stop pair 106; 107 is formed, for example, by opposing surfaces of a hook-like projection on the tensioning strip 111 and a hook-like projection 117 engaging the latter in the opposite direction, for example as a hook-in edge 117, at the end 107 of the ring element 31.

In a preferred embodiment, a site of engagement via the adjusting means 109 at the relevant end 106; 107 is straight or at least has a deviation of no more than 5°, viewed in the circumferential direction, at a point at which a tangent lying against the circumference of the cylinder inner body 32 runs parallel to the adjusting direction of the adjusting means 109. This makes it possible in the small adjustment range here that the end 106; 107 pulled by the adjusting means 109 is essentially acted upon tangentially with a force and thus radial deformation is avoided, as may occur due to a direction of force deviating from the tangent.

Although in principle also realizable in other ways, the adjusting means 109 are preferably formed by a threaded drive 113, 114, for example a threaded rod 113, in particular a screw 113, which is to be braced at the tensioning strip 111 and can be actuated manually, for example, and a corresponding thread 114, for example a threaded bushing 114, which is formed directly in the end region of the ring element 31 or preferably in a tensioning means 116 engaging on the ring element 31 and variable in position via the threaded drive 113, 114 in the adjustment direction of the threaded drive 113, 114, with the tensioning means 116 being configured and arranged so as to cooperate with the relevant end 106 via a pair of stops acting in the circumferential direction. The pair of stops is formed, for example, by opposing surfaces of a tensioning means 116 designed as a pull strip 116 and a hook-like projection 118 receiving the pull strip 116, for example as a hook-in edge 118, on the ring element 31.

In an advantageous refinement, viewed in a cross-section extending perpendicular to the cylinder axis, at least one part of the pull strip 116 is lowered into a cut-out 122 or recess 122 in the tensioning strip 111 having a corresponding shape and cross-section in such a way that movement of the tensioning strip 116 along the adjustment direction guided by the recess 122 is ensured.

In principle, a respective tensioning strip 111 and/or a respective associated tensioning means 116 can be provided for each ring element 31 to be fastened. In a preferred embodiment, however, a tensioning strip 111 extending, viewed in the axial direction of the cylinder 26, over multiple or all of the carrier elements 31 arranged on the cylinder inner body 32, and/or a tensioning means 116 extending, viewed in the axial direction of the cylinder 26,

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over multiple or all of the carrier elements **31** arranged on the cylinder inner body **32**, are provided. In this case, there is no longer any need for a fixed numerical or spatial assignment of adjusting means **109** or screw drives **113**, **114** to be assigned to a ring element **31**. The fastening device can be retained, irrespective of the number and position of the ring elements **31** with which a continuous or possibly split tensioning device **108** cooperates to clamp them. In particular, an above-mentioned tensioning device **108** without a pull strip **116**, i.e., with adjusting means **109** engaging directly in the ring element **31**, would also be less suitable for continuous positionability, since the possible positions depend on the hole spacing for the threaded rods **113**.

The tensioning strip **111** can be arranged and designed in such a way that it simultaneously forms the base support of a single-part or multi-part gripper bar. For example, bearings **121** supporting a gripper shaft **119** are arranged on the tensioning strip **111** forming the base support.

In a preferred embodiment, such a cylinder **26** is an integral part of an above-mentioned machine **01** and/or is particularly advantageous in conjunction with one or more aspects for the adjustability of individual magnetic elements **24** on respective magnetic element carriers **37** in the axial and/or circumferential direction and/or for the formation of above-mentioned action units **36** with respective magnetic and suction elements **24**; **34** and/or the clamping of individual magnetic elements **24** or their mounts **28** or magnetic element carriers **37** on the ring element **31**.

Basically independent of an arrangement of the magnetic element **24** in an above-mentioned modular unit **36** and/or of an above-mentioned adjustability in the axial direction and/or of an above-mentioned adjustability in the circumferential direction and/or of an above-mentioned clamping device for clamping the magnetic elements **24** or mounts **28** or modular units **36**, and/or of an above-mentioned design for the fastening of ring elements **31** with a tensioning device, but preferably in conjunction with one or more of the above-mentioned advantageous embodiments, in a preferred embodiment, the cylinder **26**, in particular magnetic cylinder **26**, is designed with suction openings **42** and corresponding line connections to a suction air source or air pressure sink so that the suction openings **42** only have a line connection to the suction air source or air pressure sink when a defined or definable rotation angle phase is run through or crossed over, i.e. a defined or definable angular range around the axis of rotation of the cylinder **26** of less than 360° , wherein this angular range or this rotation angle phase is preferably variable in its position and/or size in the circumferential direction. A line interface **123**, in particular a rotary union **123**, is provided, via which, when a vacuum pressure is present on the inlet side, vacuum pressure can be applied to the or a part of the suction openings **42** provided on the circumference of the cylinder **26** when running through this rotation angle phase of the cylinder **26**, also referred to as an active rotation angle phase, for example due to the presence of vacuum pressure, wherein the rotary union **123** is preferably adjustable with respect to a position and/or size of an angle sector $\Delta\phi$ determining the active rotation angle phase for conducting through the suction air, hereafter referred to as the passage angle sector $\Delta\phi$ for short.

Suction openings **42** are provided on the circumference of the cylinder **26** over the circumference or at least one circumferential section, which extends, for example, over at least half of the cylinder circumference, in individual segments Δ_x , i.e. partial sections of the circumference or of the above-mentioned circumferential section (see, for example, FIG. 5), in such a way that suction openings **42** or groups of

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suction openings **42** in at least one of multiple consecutive segments Δ_x in the circumferential direction of the cylinder **26**, for example angle or ring segments Δ_x , can advantageously be supplied with vacuum pressure or suction air independently of an adjacent segment Δ_x in the circumferential direction, i.e. can be connected to or disconnected from the suction air source or air pressure sink. For this purpose, the angle segments Δ_x in the cylinder **26**, which can be acted upon separately, can be supplied with vacuum pressure or suction air, for example through separate line paths. This can be realized, for example, via a structure that is accordingly segmented in the circumferential direction for conducting through the suction air, for example in an advantageous embodiment with individual segments **45** for the segmented conduction of suction air, for example in the above embodiment with groups of suction openings leading into separate chambers **55**.

For the conduction of the suction air, the suction openings **42** or groups have a line connection via corresponding lines, channels and/or chambers in the cylinder **07** to the above-mentioned rotary union **123**, which has a cylinder-fixed and co-rotating part **124**, for example hereafter also referred to as rotor **124** of the rotary union **123**, and a part **126** that is fixed during normal operation, for example hereafter also referred to as stator **126** of the rotary union **123**. In the mounted state, the latter cooperates with the rotor **124** for conducting through the suction air in a passage angle sector $\Delta\phi$, in particular based on the position around the axis of rotation of the rotor **124** or cylinder **26**, and has a line connection on its suction side, i.e. with a suction-side outlet of the rotary union, downstream with a suction air source as defined above or can be brought into line connection with such a source.

Viewed in the circumferential direction of the cylinder **07**, the suction openings **42** or groups of suction openings **42** of multiple or all successive angle segments Δ_x , for example via corresponding segments **45** for the conduction, have a line connection segment by segment via separate line paths with respective openings, for example channel openings **127** of the rotating part **124**, i.e. the rotor **124** of the rotary union **123**, which is designed in particular with multiple channels, and can thus be supplied with suction air separately from one another, segment by segment, via the corresponding channel openings **127** and the relevant line path. The channel openings **127** are preferably arranged eccentrically, in particular concentrically around the axis of rotation of the cylinder **26** or an axis coinciding with the axis of rotation of the cylinder **26**, and are provided in the same sequence and/or number as the corresponding angle segments Δ_x or associated segments **45** having the same sequence or number when viewed in the circumferential direction.

The rotary union **123** is preferably non-rotatably arranged with its cylinder-fixed part **124** at the end face on a cylinder journal **128** formed by a lateral shaft end **128** of the shaft **32** or otherwise, in particular laterally from the cylinder end face and/or outside the cylinder envelope. The channel openings **127** could in principle be provided on the circumference of the rotor **124** and cooperate with recesses on the inner circumference of a stator **126** surrounding the rotor **124** or vice versa. In the preferred embodiment shown here, however, the channel openings **127** are provided on the end face and in the above-mentioned manner eccentrically and in particular concentrically to the axis on the rotor **124** and cooperate on the end face with one or more eccentrically arranged recesses **129** of the stator **126** conducting through the suction air or the vacuum. The latter is or are connected or connectable on the suction side via corresponding line

paths to the suction air source or air pressure sink and are arranged eccentrically in such a way that they can be at least intermittently brought at least partially into alignment with each of the channel openings 127 of the angle segments Δ_x or segments 45 to be supplied with suction air by relative rotation of the rotor 124 and stator 126 about the axis of rotation of the cylinder 26 or an axis coinciding with the axis of rotation of the cylinder 26. When the channel opening 127 is at least partially aligned with the or a recess 129, the suction openings 42 in the relevant angle segment Δ_x or corresponding segment 45 are or can be supplied with suction air or vacuum pressure.

In a preferred embodiment, the stator 126 that does not rotate during normal operation can be adjusted in its rotational position about the axis of rotation of the cylinder 26 or an axis coinciding therewith, so that the angular position of the recess 129 or recesses 129 cooperating with the channel openings 127 for conduction can be varied about this axis of rotation. If a recess 129 or group of recesses 129 on the stator 126 extending over a circular arc segment is now opposite the group of channel openings 127 that are arranged eccentrically on the rotor 124 and can be brought into alignment in the above manner by relative movement, the rotation angle position in which a channel opening 127 enters in an area of alignment with the or a recess 129 can be varied by rotating the stator 126. In this form of the stator 126 or of the rotary union 123 comprising such a stator 126, the angular position of leaving the alignment is also varied to the same extent when the stator 126 is rotated. By adjusting the angular position of the recess 129 or recesses 129 cooperating, for conduction, with the channel openings 127 by a rotation of the stator 126, assuming a corresponding arrangement of the angle segments Δ_x or segments 45 to be supplied with suction air or vacuum pressure via the channel openings 129 in terms of sequence and/or number, the position of the passage angle sector $\Delta\phi$ and thus the position of the above-mentioned active rotation angle phase about the axis of rotation of the cylinder 26 can be varied, within which suction openings 42 have a line connection to the suction air source or air pressure sink when the cylinder 26 rotates while the rotation angle phase is being run through.

In a particularly advantageous embodiment, in which, for example, sheet travel of the substrate sheets 02 to be transported can be even further improved, the stator 126 is designed in multiple parts in such a way that a free cross-section for the passage of suction air through the recess 129 or recesses 129 is variable via a relative rotational movement between a first stator part 131, which, for example, comprises the recess 129 or recesses 129, and a second stator part 132, which, for example, depending on the relative rotational position relative to the first stator part 131, closes off a variable part of the recess 129 or recesses 129 of the first stator part 131 against the passage of suction air. The relative rotational movement takes place, for example, about an axis of rotation parallel to, in particular in alignment with, the axis of rotation of the cylinder 26, preferably by rotation of the second stator part 132.

In this multi-part, in particular two-part design, the rotary union 124 is designed in such a way that a start of the active rotation angle phase, i.e. the rotation angle for the start of suction when the cylinder 26 rotates, and an end of the active rotation angle phase, i.e. the rotation angle for the end of suction when the cylinder 26 rotates, can be selected or adjusted independently of one another over at least a respective adjustment range by a positioning of the two stator parts 131; 132 in each case separately and relative to one another.

In a depicted and preferred first variant (see, for example, FIG. 19 to FIG. 21), the first stator part 131 comprises a recess 129 which preferably extends over at least part of its length in a circular segment-like manner in the circumferential direction and is arranged directly adjacent to the rotor 124 as viewed in the axial direction in such a way that the channel openings 127 of the rotor 124 and an end-face opening of the recess 129, which is in particular in the form of a circular segment, are located at least partially directly opposite one another with an accordingly suitable relative position. The second stator part 132, provided in this variant embodiment at the end face on the side of the first stator part 131 facing away from the rotor 124, comprises a cover element 133, for example a circular disk sector 133, which, for example, is arranged within a closed circular ring and, depending on the rotational position, covers a variable part of the recess 129 or of the circular ring segment-like part thereof on the side facing away from the rotor 124, and a blocking element 134, which is arranged on one side of the cover element 133 as viewed in the direction of rotation and which, based on the cross-sectional profile of the recess 129 perpendicular to the circular ring segment-like longitudinal extension, engages in a shape-complementary manner in the circular ring segment-like recess 129 or at least its ring segment-like section in such a way that it suppresses the free passage of suction air between the covered and the free parts of the recess 129 within the recess 129. This shall also include designs in which residual quantities still pass through or can pass through, for example due to tolerances. Depending on the relative rotational position between the first and second stator parts 131; 132, a variable part of the recess 129 on the suction side is covered to prevent the passage of suction air in the axial direction and a significant passage of suction air from the covered part of the recess 129 is suppressed by the blocking element 134 engaging in the recess 129 in the above sense. Accordingly, suction air only still passes through the stator 126, at least to a notable extent, in the alignment area of the free recess 129 and thus through the rotary union 123. By varying the circumferentially effective size of the free part of the recess 129 that cooperates with the channel openings 127 for conduction, it is possible, in addition to the above-mentioned variation of the position of the above-mentioned rotation angle phase about the axis of rotation of the cylinder 26, within which suction openings 42 or underlying segments 45 have a line connection with the suction air source or air pressure sink when the cylinder 26 rotates while the rotation angle phase is being run through, to also vary the size of the passage angle sector $\Delta\phi$ and thus the above-mentioned active in the circumferential direction by relative rotation between the first and second stator parts 131; 132.

In an alternative variant embodiment (not shown here), the second stator part 132 is arranged with the cover element 133 directly adjacent to the rotor 124, and the first stator part 131 is arranged on the side facing away from the rotor 124, and the provided cover element 133 is, for example, a circular ring segment 133 or a circular disk sector 133 arranged within a circular ring, wherein, however, due to the arrangement of the cover element 133 on the side of the rotor 124, an above-mentioned locking element 134 can be omitted.

For adjusting the stator 126 or, if designed in two parts, the first stator part 131 including the at least one recess 129, adjusting means 136, 141, for example an adjusting shaft 136, for example in the form of a hollow shaft 136 or sleeve 136, which is non-rotatably connected to the stator 126 or the first stator part 131, and an adjusting lever 141, via

which, for example, the adjusting shaft **136** can be pivoted, are provided. By pivoting the stator **126** or the first stator part **131**, the angular position of the at least one recess **129** and thus the position of the passage angle sector $\Delta\phi$ or of the active rotation angle phase can be varied in the aforementioned manner, in which openings are supplied with vacuum pressure or suction air when the cylinder **26** rotates. In order to improve reproducibility, for example, an indicator **148** connected to the adjusting lever **141** can be provided, which cooperates with a fixed scale **149**. In the case of a one-piece stator **126**, for example, the suction air can be conducted through the adjusting shaft **139**, which is designed as a hollow shaft **136**. In order to prevent inadvertent displacement, a clamping element **144**, for example in the form of a screw nipple, can be provided, by means of which the adjusting lever **141** can be optionally secured to prevent displacement.

If the stator **126** comprises two stator parts **131**, **132** as described above, further adjusting means **137**, **139**, for example a further adjusting shaft **137** connected to the second stator part **132**, for example a hollow or slotted shaft **137** conducting the suction air from the suction-side outlet of the rotary union **123** formed by the stator **126** and the cooperating rotor **124**, which extends, for example, inside the above-mentioned hollow shaft **136**, and, for example, a further adjusting lever **139**, via which, for example, the adjusting shaft **137** can be pivoted, are provided. By pivoting the second stator part **132** while the first stator part **131** remains idle, the length of the free cross-section of the recess **129** for the passage in the circumferential direction and thus the size of the passage angle sector $\Delta\phi$ of the rotary union **123** or of the active rotation angle phase, in which openings are supplied with vacuum pressure or suction air when the cylinder **26** rotates, can be varied in the aforementioned manner. In order to improve reproducibility, for example, an indicator connected to the adjusting lever **139** via a sleeve or rod **146** can be provided, which cooperates with a fixed scale **147**. In order to prevent inadvertent displacement of the adjusting lever **139**, a clamping element **143**, for example in the form of a screw nipple, can be provided, by means of which the adjusting lever **141** can be selectively secured to prevent displacement. A fixed sleeve can be provided between the two adjusting shafts **136**; **137** to support them.

The line connection which conducts the suction air through to the rotary union **123**, for example a relevant adjusting shaft **136**; **137**, has a line connection in the area of its end, for example via an air chamber, with a suction air line **142**, which in turn has a line connection with the suction air source or air pressure sink.

In the mounted state, the cylinder **26** is accommodated, for example, in one or more bearing shells **151**, which are rotatably mounted in the side frame via a radial bearing (not visible here), wherein the rotor **124** (see, for example, FIG. 4 or FIG. 15) and the stator **126** (see, for example, FIG. 19 or FIG. 20) cooperate on the end face in the manner described above for the conducting through of the suction air.

The above-mentioned embodiment of a cylinder **26** with suction openings **42** or groups of suction openings **42**, which only have a line connection with the suction air source or air pressure sink when an above-mentioned active rotation angle phase is run through or crossed over, i.e. a defined or definable angular range about the axis of rotation of the cylinder **26** of less than 360° , wherein the passage angle sector $\Delta\phi$ or the active rotation angle phase is preferably variable in its position and/or size in the circumferential direction, is described above in connection with a preferred

embodiment of the cylinder **26** as a magnetic cylinder **26**, but can also be applied without contradiction to other cylinders **12**; **17**; **153**, in particular transport cylinders **12**; **17**; **153**, in further particularly advantageous embodiments that act as transport cylinders **12**; **17**; **153** and, for example, comprise holding means **33** for gripping the cylinders **12**; **17**; **153**. For example, holding means **33** for gripping substrate sheets **02** around the circumference of a sheet-treating- and/or sheet-processing machine **01**, in particular a securities machine **01**, for example printing machine **01** or in particular a securities printing machine **01**.

The machine **01** can be designed in the above manner with, for example, a magnetic cylinder **26** designed in the above manner. In addition or instead, however, the impression cylinder **17**, which acts as a transport cylinder **17** in the case of a sheet-format substrate **02**, and/or a cylinder **152**, for example an inspection cylinder **152**, which acts as a transport cylinder **152** and cooperates with an above-described sensor device **153**, for example camera **153**, in particular line scan camera **153**, can be designed, in an above-mentioned embodiment, with suction openings **42** or groups of suction openings **42**, which only have a line connection with the suction air source or air pressure sink when an above-mentioned active rotation angle phase is run through or crossed over, i.e. a defined or definable angular range about the axis of rotation of the cylinder **26** of less than 360° , wherein the rotation angle phase can preferably be varied in its position and/or size in the circumferential direction as described above based on the example of the magnetic cylinder **26**. All these cylinders **26**; **17**; **152** have in common that they are an integral part of a device acting on the substrate sheet **02** or inspecting the substrate sheet **02** and require a particularly secure support and an optimized sheet transfer behavior.

In principle, the above-mentioned adjustability of an active rotation angle phase in position and/or size, due to the optimizable transfer behavior, can also be advantageous for pure transport cylinders **12**, which do not have to fulfill any other function in addition to the transfer of substrate sheets.

In a particularly advantageous refinement, such a rotary union **123** is provided for each end face of the cylinder **26**; **17**; **152**; **12**, for example in order to apply or be able to apply vacuum pressure to the relevant suction openings **42** or groups of suction openings **42** from both sides in the active rotation angle phase. Vacuum pressure can be applied to the chambers **55** from both sides or the chambers can be divided in the axial direction and vacuum pressure can be applied separately.

Although the disclosure herein has been described in language specific to examples of structural features and/or methodological acts, it is to be understood that the subject matter defined in the appended claims is not necessarily limited to the specific features or acts described in the examples. Rather, the specific features and acts are disclosed merely as example forms of implementing the claims.

The invention claimed is:

1. A cylinder (**26**; **17**; **152**; **12**) for a sheet-treating and/or sheet-processing machine (**01**), the cylinder (**26**; **17**; **152**; **12**) comprising a holding means (**33**) around a circumference thereof, by which a substrate sheet (**02**) to be conveyed over the cylinder (**26**; **17**; **152**; **12**) can be picked up at a leading end thereof and can be or is held during a rotation of the cylinder (**26**; **17**; **152**; **12**) over a rotation angle range between when the substrate sheet (**02**) is received and when the substrate sheet (**02**) is transferred downstream, the cylinder (**26**; **17**; **152**; **12**) further comprising a rotary union (**123**), which comprises a rotor (**124**) rotating along with the

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cylinder (26; 17; 152; 12) which, at an end face, is non-rotatably arranged at a shaft end (128) of a shaft (32) driving the cylinder (26; 17; 152; 12) or at an end-face cylinder journal (128), as well as a stator (126) that does not rotate during regular operation, the rotary union (123) configured to enable vacuum pressure to be applied to suction openings (42) or groups of suction openings (42) provided around the circumference of the cylinder (26; 17; 152; 12) while an active rotation angle phase of the cylinder (26; 17; 152; 12) about an axis of rotation (R) thereof is being run through, characterized in that

the rotary union (123) can be set in terms of a size of a passage angle sector (Df) determining a size of the active rotation angle phase and/or in terms of a position of the passage angle sector (Df) determining a position of the active rotation angle phase; and

that the stator (126) is configured as a multi-piece stator (126) comprising a first stator part (131) and a second stator part (132), which can both be adjusted, in terms of a rotational position thereof with respect to one another and relative to an axis of rotation, about the axis of rotation (R) of the cylinder (26; 17; 152; 12) or an axis coinciding therewith.

2. The cylinder according to claim 1, characterized in that the rotary union (123) has a multi-channel design on a side of the rotor (124), including line connections which originate on a cylinder side from channel openings (127) of the multi-channel rotor (124) leading to different suction openings (42) or groups of suction openings (42) located one behind the other in a circumferential direction.

3. The cylinder according to claim 2, characterized in that the stator (126) comprises a recess (129) or a group of recesses (129) used for passage of suction air, which, depending on a rotational position of the rotor (124) rotating along with the cylinder (26), is in alignment with a changing part of the channel openings (127) of the rotor (124).

4. The cylinder according to claim 3, characterized in that the stator (126) comprising the recess (129) or group of recesses (129), for varying the position of the passage angle sector (Df) and thus the position of the defined rotation angle phase and/or for varying the position of the channel openings (127) which at the same time are in alignment with the recess (129) or group of recesses (129), and the suction openings (42) or groups of suction openings (42) having a line connection thereto, can be pivoted about the axis of rotation (R) of the cylinder (26) or an axis coinciding therewith.

5. The cylinder according to claim 3, characterized in that the stator (126) is configured as a multi-piece stator (126) comprising a first stator part (131) including a recess (129) or a group of recesses (129), and a second stator part (132) comprising a cover element (133), the first and second stator parts (131, 132), for varying the size of the passage angle sector (Df) and thus the size of the defined rotation angle phase and/or for varying a size of the free passage cross-section through the recess (129), and thus a portion of the channel openings (127) that are simultaneously in alignment with this recess (129), are arranged so as to be variable in terms of a relative rotational position thereof.

6. The cylinder according to claim 5, characterized in that the second stator part (132) can be pivoted relative to the first stator part (131) and/or comprises a blocking element (134) engaging in the recess of the first stator part (131), which separates a part of the recess (129) not covered by the cover element (133) from a covered part of the recess (129).

7. The cylinder according to claim 2, characterized in that the stator (126), for conducting through suction air, is

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configured with a recess (129) or a group of recesses (129) in such a way that, during a rotation of the cylinder (26; 17; 152; 12) and of the rotor (124) of the rotary union (123) which rotates along with the cylinder, always only that part of the channel openings (127) has a line connection to a free cross-section of the recess (129) or groups of recesses (129) of the stator (126) which is connected via corresponding line paths to suction openings (42) or groups of suction openings (42) which are located within the rotation angle phase of the cylinder (26; 17; 152; 12) about the axis of rotation (R) thereof of less than 360°.

8. The cylinder according to claim 1, characterized in that the position and size of the passage angle sector (Df) determining the size and the position of the active rotation angle phase and/or a beginning and an end of the active rotation angle phase, across at least a respective adjustment range, can be selected or set by a positioning of the two stator parts (131; 132) with respect to one another and the respective angular position thereof assumed about the axis of rotation or an axis coinciding therewith.

9. The cylinder according to claim 1, characterized in that the second stator part (132) can be pivoted by way of adjusting means (137, 139) relative to the first stator part (131) and/or about the axis of rotation (R) of the cylinder (26) or an axis coinciding therewith and/or that the first stator part (131) can be pivoted by way of adjusting means (136, 141) about the axis of rotation (R) of the cylinder (26) or an axis coinciding therewith and/or relative to the second stator part (132).

10. The cylinder according to claim 1, characterized by the design thereof as a magnetic cylinder (26) comprising a number of $n \times m$ (where $n, m \geq 0$) magnetic elements (24), arranged in a matrix-like manner, in a region of an outer circumference thereof, which are arranged in axially parallel extending rows and in columns extending in a circumferential direction.

11. The cylinder according to claim 10, characterized in that a plurality or all of the magnetic elements (24) of the columns extending in the circumferential direction are arranged at or on a respective carrier element (31), which is open or closed in a ring-like manner and accommodated on a cylinder shaft (32), the ring-like carrier element (31), viewed in the circumferential direction, comprising a plurality of chambers (55) one behind the other, which each, independently of one another, have a line connection to the rotor (124) of the rotary union (123) and to at least one group of suction openings (42) joining at the circumference via corresponding line paths.

12. The cylinder according to claim 1 characterized in that a plurality of channels (48) leading into a cylinder interior are provided in the shaft (32) carrying the cylinder (26; 17; 152; 12) or in the end-face cylinder journal (128), which have a line connection to end-face channel openings (127) of the rotor (124).

13. A machine (01) for treating and/or processing sheet-format substrate (02), comprising a substrate infeed (13), at least one printing mechanism (04), by which substrate (02) guided on a transport path through the machine (01) is and/or can be printed at least on a first side in a matrix-like manner with multiple-ups (09) having a number m of columns and a number n of rows, a product receiving system (22), by which processed substrate (02) can be combined into bundles, as well as at least one transport cylinder (26; 17; 152; 12) provided in a substrate path between the substrate infeed (13) and the product receiving system (22),

wherein the transport cylinder (26; 17; 152; 12) is configured according to a cylinder (26; 17; 152; 12) according to claim 1.

14. The machine according to claim 13, characterized in that the transport cylinder (26) according to the cylinder (26) 5 according to claim 1 is provided in the substrate path, which is designed as a magnetic cylinder (26) of an alignment device (07) provided between the printing mechanism (04) and the product receiving system (22) for aligning magnetic or magnetizable particles (P), which in a region of an outer 10 circumference thereof comprises a number of $n \times m$ (where $n, m \geq 0$) magnetic elements (24) arranged in a matrix-like manner, which are arranged in axially parallel extending rows and in columns extending in a circumferential direction, and/or that the transport cylinder (17) according to the 15 cylinder (17) according to claim 1 is provided in the substrate path, which, as an impression cylinder (17) of the printing mechanism (04), serves as a counterbearing for a printing mechanism cylinder (14) and, together with the counterbearing, forms a printing nip (11) and/or that the 20 transport cylinder (152) according to the cylinder (152) according to claim 1 is provided in the substrate path, which, as an inspection cylinder (152), provides support to a conveyed substrate (02) and, as an integral part of an inspection device, cooperates with a sensor device (153) 25 directed at the substrate path.

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